REVIEW ARTICLE

From Superspace Model Sets to Information-Theoretic Time: Readout-Based Space-Time Reconstruction with Quantitative Resolution and Capacity Laws

Haobo Ma[1*]

[1]AELF PTE LTD., #14-02, Marina Bay Financial Centre Tower 1, 8 Marina Blvd, Singapore, 018981, Singapore.

Corresponding author(s). E-mail(s): auric@aelf.io;

Abstract

We develop a geometry-first framework for spacetime reconstruction in which time and causality emerge from superspace model sets and finite-resolution readout protocols, without presupposing a continuum background. The construction proceeds along a self-contained dependency chain: (i) cut-and-project schemes furnish geometric primitives; (ii) windowed readout maps internal degrees of freedom to symbolic data streams, with quantitative resolution laws (Tanaka–Stokes scan-error representation, boundary-dimension exponent); (iii) Fold-type stabilization compresses raw words into golden-mean constrained languages, with worst-case gauge anomaly of full word length and typical density $\mathbf{4/9}$; (iv) information-theoretic time, defined via entropy-rate refinement (SMB/KS framework), satisfies a KL-divergence ledger identity and locks onto a canonical $\mathbf{log}\,\boldsymbol{\varphi}$ time scale; (v) a double-budget capacity law decomposes addressability into synchronization and collision budgets, grounding a causality interface. We further prove seven structural results including an exact clock formula with Parry rigidity (the golden-mean Parry measure is the unique zero-jitter Markov clock), an explicit variance formula for non-Parry clocks, a sub-golden no-go theorem, and a capacity–entropy duality. All numerical fingerprints are generated by a single reproducible script. We conclude with falsifiable predictions and open problems for connecting superspace readout to classical and quantum spacetime geometry.

Keywords: cut-and-project; superspace; model sets; symbolic dynamics; entropy rate; golden ratio; Zeckendorf stabilization; gauge anomaly; causality; spacetime reconstruction

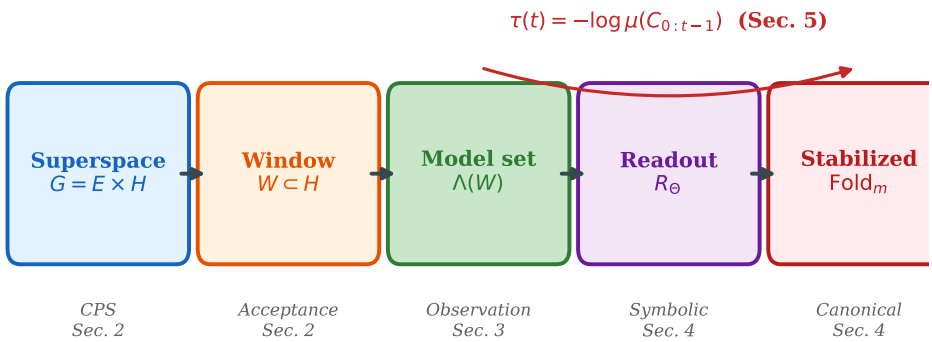

Fig. 1 Schematic overview of the readout pipeline. A superspace lattice (Section 2) is sliced by a window to produce a model set; the model set is sampled by a finite-resolution readout (Section 3); raw symbolic data are stabilized into Zeckendorf canonical form (Section 4). The information-theoretic time $\tau(t) = -\log \mu(C_{0:t-1})$ (Section 5) quantifies entropy accumulation along the stabilized record.

1 Introduction

The topical collection *Geometry of Classical and Quantum Space-Times* foregrounds a recurring theme in gravitational physics: geometry is not merely a stage on which physics happens, but a structure that may itself be reconstructed, constrained, or even *defined* by operational and information-theoretic considerations. This review is written in that spirit, and it is intentionally *geometry-first*: we begin from well-posed geometric objects and dynamical systems and then introduce an explicit *readout protocol* that turns those objects into finite-resolution data streams. The central organizing idea is that "spacetime" can be approached as an emergent interface—a stable quotient of a richer geometry under a constrained observation map—rather than as a primitive background.

1.1 Scope and guiding separation of layers

Throughout, we maintain a strict separation between three layers:

(i) **Object layer** (geometry/dynamics): superspace embeddings, lattices, torus translations, cut-and-project schemes, model sets, and their invariant measures.

(ii) **Protocol layer** (readout): finite-resolution windowing, coarse-grainings, symbolic codings, canonicalization/stabilization maps, and the induced fibers of unresolved degrees of freedom.

(iii) **Interpretation layer** (physics): how the protocol-level objects can be mapped to familiar spacetime notions (time, causality, observables), and where such mappings should *not* be over-claimed.

This separation prevents category errors (e.g. treating a protocol artifact as an ontological claim) while keeping the mathematical backbone auditable. The overall architecture is summarized in Figure 1.

1.2 Main contributions

As a review article, the primary contribution is synthesis. Nevertheless, we aim to contribute *structure with proofs* rather than mere exposition. Beyond collecting and organizing known material, this paper establishes the following new results:

C1. Quantitative resolution laws (Section 3). We prove a Tanaka–Stokes representation (Theorem 3.3) expressing scan error as a martingale local-time accumulation, and derive a boundary-dimension exponent (Theorem 3.4) that governs the exponential rate at which readout distinguishes events. A volume–boundary scaling law (Theorem 3.5) provides matching upper and lower bounds.

C2. Stabilization with gauge anomaly control (Section 4). We prove that the Fold canonicalization map admits a worst-case gauge anomaly of full word length (Theorem 4.3), that the typical anomaly density converges to 4/9 under CLT scaling (Theorem 4.4), and that the sequence-level stabilization factor Φ_m is a topological conjugacy for all $m \geq 3$ with a computable inverse of memory $m - 1$ (Theorem 4.5). These results quantify the non-trivial cost and benefit of canonicalization.

C3. Observable algebra non-expansion under recursion (Section 4). We prove that recursive addressing—constructing new concepts from readout of old ones—does not enlarge the visible σ-algebra (Corollary 4.8), formalizing the principle that observation cannot create information beyond what the object layer provides.

C4. KL-divergence ledger and entropy-rate time (Section 5). We establish an exact identity (Theorem 5.2) decomposing Shannon entropy under stabilization into a stable-type term, a fiber degeneracy term, and a KL residual measuring departure from fiber-uniform distributions. This yields a canonical $\log \varphi$ time scale (Proposition 5.5) locked by the Parry measure on the golden-mean shift.

C5. Double-budget capacity law and causality interface (Section 6). We prove a capacity inequality (Theorem 6.1) decomposing addressability failure into synchronization timeout and collision probability, providing quantitative bounds on how many events can be simultaneously distinguished at given resolution and scan depth.

C6. Optimality of the golden-ratio slope (Section 7). We prove that among constant-type continued-fraction slopes (metallic means), the golden ratio uniquely minimizes the leading coefficient of the deterministic TV certificate for stabilized-type histograms (Corollary 7.3).

1.3 What is not claimed

To reduce the risk of over-interpretation, we state explicitly what is *not* claimed:

- The constructions reviewed here do not derive Einstein dynamics, reconstruct a Lorentzian metric, or compute curvature invariants from first principles.
- The word "superspace" is used exclusively in the quasicrystal/model-set sense of a cut-and-project embedding space $G = E \times H$ (Section 2). It is *not* Wheeler's superspace

(configuration space of geometrodynamics) and does not involve Grassmann variables (supersymmetric superspace).

- Protocol-level time and causality depend on protocol parameters Θ. Identifying protocol-invariant structures (or universality classes) is essential before making strong physical claims.

1.4 Roadmap

Section 2 introduces superspace model sets as geometric primitives. Section 3 formalizes finite-resolution readout and proves quantitative resolution laws. Section 4 develops stabilization maps, gauge anomaly theorems, and the non-expansion principle. Section 5 defines information-theoretic time via refinement and entropy rates, establishing the KL-ledger identity. Section 6 discusses the double-budget capacity law and causality interfaces. Section 7 lists falsifiable fingerprints with numerical demonstrations. Section 8 addresses limitations, connections to established GR/QG themes, and presents seven new structural results: a scan-error–fiber bound (Proposition 8.1), an exact clock formula with Parry rigidity characterization (Theorems 8.2 and 8.3), an explicit asymptotic variance formula for non-Parry clocks (Proposition 8.4), a sub-golden no-go theorem (Proposition 8.5), a fiber-entropy–window-regularity estimate (Proposition 8.6), and a capacity–entropy duality (Proposition 8.7).

1.5 Positioning within spacetime reconstruction programs

Spacetime reconstruction appears across a broad range of classical and quantum gravity approaches. We briefly situate the present viewpoint relative to several established themes:

- **Causal-set and causal-structure-first approaches.** In causal set theory, causal order and discreteness are fundamental, with continuum Lorentzian geometry expected to emerge in appropriate limits [24, 25]. We do not postulate a causal order at the object layer; instead, we define a protocol precedence relation induced by refinement (Section 6) and discuss what would be needed to relate it to Lorentzian causal orders.
- **Quantum geometry via operator algebras.** Approaches such as loop quantum gravity emphasize that geometric information is encoded in noncommuting operators and their spectra [27, 28]. Our "readout as operator/protocol" language resonates with this viewpoint, but we remain agnostic about quantization.
- **Spectral and information-theoretic reconstructions.** The entropy-rate time defined here (Section 5) belongs to this family, but it is grounded in the SMB/KS theorem chain and tied explicitly to a readout protocol with quantitative error control. Discrete approaches such as Regge calculus [38] and causal dynamical triangulations [39, 40] reconstruct geometry from combinatorial building blocks; our framework shares the emphasis on discrete primitives but replaces simplicial complexes with symbolic data streams.
- **Superspace and higher-dimensional embeddings.** In quasicrystal theory, higher-dimensional periodic embeddings provide a mature technology for generating

aperiodic patterns with controlled harmonic structure [5]. Noncommutative geometry [41] provides complementary spectral tools, and the cohomological invariants of tiling spaces [42–44] offer a topological perspective. We import the cut-and-project technology as a geometric backbone and then ask how protocol-limited observation turns it into "spacetime-like" notions.

The practical message is that this article is not a proposal of a new quantum gravity model. Rather, it is a structured review of a reusable interface: *superspace geometry + finite-resolution readout* as a disciplined generator of symbolic data streams, stable types, fiber structures, and entropy-controlled arrows. This interface can be used as a building block inside different reconstruction programs, provided its assumptions and limits are kept explicit.

2 Superspace model sets as geometric primitives

Superspace methods provide a standard geometric language for quasi-periodic structures and nonperiodic order, originating from the mathematical theory of Meyer sets [1, 3] and given physical impetus by the experimental discovery of quasicrystals [47, 48]. For this review, the key point is simple: aperiodic patterns in the physical space E are generated by a periodic lattice in $G = E \times H$ together with a protocol-specified selection in the internal space H. The cut-and-project construction formalizes this mechanism [1, 2, 5, 45].

2.1 Cut-and-project schemes

We fix finite-dimensional real vector spaces

$$E := \mathbb{R}^d \quad \text{(physical space)}, \qquad H := \mathbb{R}^n \quad \text{(internal space)},$$

and let $G := E \times H$. Let $\pi_E : G \to E$ and $\pi_H : G \to H$ denote the coordinate projections.

Definition 2.1 (Cut-and-project scheme) A *cut-and-project scheme* (CPS) consists of (E, H, Γ) where $\Gamma \subset G$ is a lattice such that:

(i) $\pi_E|_\Gamma$ is injective;

(ii) $\pi_H(\Gamma)$ is dense in H.

We write $\Gamma_E := \pi_E(\Gamma)$ and define the *star map* $^\star : \Gamma_E \to H$ by $x^\star := \pi_H(\gamma)$ where $\gamma \in \Gamma$ is the unique lattice point with $\pi_E(\gamma) = x$.

In the quasicrystal/superspace context, one often takes $d = 3$ and $d+n = 6$ (a "rank-6" superspace) to capture quasi-periodic structures with six independent reciprocal vectors [45, 46]. We keep d and n general but use "6D" language when emphasizing the GRG-motivated corpus.

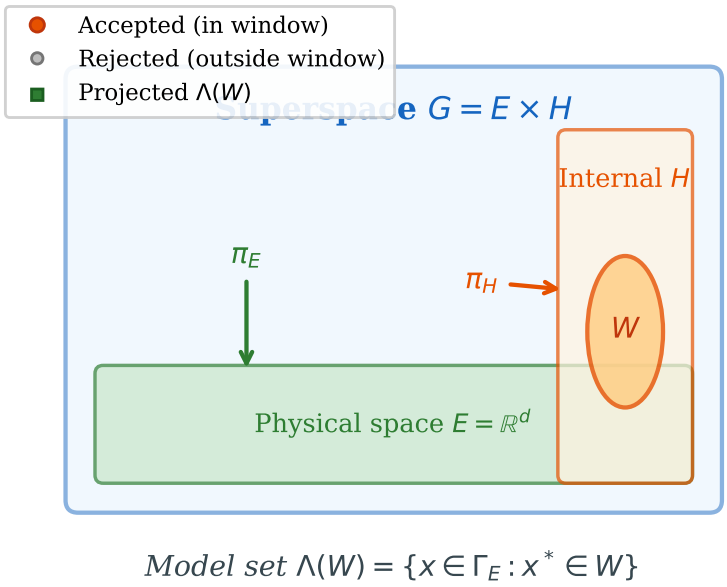

Fig. 2 Schematic of a cut-and-project scheme. Lattice points in the superspace $G = E \times H$ (blue box) are tested for membership in the window $W \subset H$ (orange ellipse). Accepted points (orange circles) project onto the model set $\Lambda(W) \subset E$ (green squares); rejected points (grey) are discarded. The star map $^\star : \Gamma_E \to H$ mediates between the physical projection π_E and the internal projection π_H.

2.2 Windows, acceptance domains, and model sets

The observational content of a CPS is extracted by selecting a compact subset $W \subset H$ (a *window* or *acceptance domain*) that specifies which lattice points are "accepted" by the slice.

Definition 2.2 (Model set) Given a CPS (E, H, Γ) and a window $W \subset H$, the associated *model set* is

$$\Lambda(W) := \{x \in \Gamma_E : x^\star \in W\} \subset E.$$

If W is compact with nonempty interior and ∂W has Haar measure zero, $\Lambda(W)$ is called a *regular model set.*

Remark 2.1 (Internal phase) *Including an internal phase $u \in H$ and considering $\Lambda(W, u) := \{x \in \Gamma_E : \ x^\star \in W + u\}$ parameterizes "observer phase" at the object layer.*

2.3 Invariant measures and typicality

Cut-and-project constructions have a natural dynamical system behind them: translations on the compact Abelian group G/Γ. Under mild hypotheses, one obtains unique ergodicity and well-defined pattern frequencies.

Proposition 2.1 (Unique ergodicity for regular model sets) *Let (E, H, Γ) be a CPS and let $W \subset H$ be a compact window with nonempty interior and $\mu_H(\partial W) = 0$. Let X_Λ be the hull of $\Lambda(W, u)$ in the local topology, equipped with the E-translation action. Then (X_Λ, E) is uniquely ergodic, and for every finite patch P the frequency* $\mathrm{freq}(P; \Lambda(W, u))$ *exists and is independent of u for μ_H-a.e. phase. [2, 5]*

2.4 Fourier modules and spectral structure

A major reason superspace model sets are attractive as geometric primitives is that they come with an explicit reciprocal-space description. The Fourier module of a model set is determined by the dual lattice $\Gamma^* \subset G^*$ and by the dual projections. For regular model sets, the diffraction measure is pure point, with Bragg peak intensities controlled by the window geometry [5, 6]:

Proposition 2.2 (Pure-point diffraction) *For a regular model set with Riemann-integrable window weight w (including $w = \mathbf{1}_W$ and smooth kernels), the diffraction measure is pure point and supported on a Fourier module determined by Γ^* and the star map. For k in this module, the Bragg intensity is proportional to $|\widehat{w}(k^\star)|^2$. [5–7]*

2.5 The torus parametrization viewpoint

The compact group G/Γ plays the role of a maximal equicontinuous factor for the translation action on the hull. The internal phase u can be identified with a point on this torus. "Varying the observer" corresponds to shifting along a compact parameter space, while the observed pattern in E is a factor determined by the window. This language is particularly convenient when readout is modeled as sampling along a one-dimensional factor: a scan is literally a translation on G/Γ projected onto a low-dimensional torus factor [8, 11, 49].

2.6 Rational approximants vs. irrational limits

Irrational orientations are never realized with infinite precision. In CPS settings this leads to a natural hierarchy of *rational approximants*: a rationally oriented strip yields periodic patterns, while the irrational limit yields a quasi-periodic model set. Concrete 6D constructions exhibit this dichotomy explicitly, including families where a rational $(1/1)$ strip produces a periodic phase and an irrational $(\tau/1)$ strip produces an icosahedral quasicrystalline phase [50].

2.7 What superspace does not imply

The superspace $G = E \times H$ is a representation space for controlled cut-and-project geometry. It is *not* an assumption of physical extra dimensions, and it does not by itself imply Einsteinian dynamics. We use superspace to supply an explicit, spectrally convenient object layer, while readout, stabilization, time, and causality are treated as protocol-defined interfaces whose physical interpretation requires additional assumptions.

3 Finite-resolution readout and quantitative resolution laws

Model sets provide a geometric mechanism for generating an aperiodic pattern in physical space. To speak about *observation*, we must specify how a finite agent extracts information from the underlying geometric object.

3.1 Observer parameters

At the protocol layer we package the observer's choices into a parameter tuple

$$\Theta := (\text{orientation},\ \text{internal phase},\ \text{window},\ \text{resolution kernel},\ \text{scan rule}).$$

The parameter Θ determines: which degrees of freedom are treated as "internal" and hence unobserved except through projection; which subset of internal space counts as "accepted" (the window); how sharply the window boundary is resolved (finite resolution); and how the internal phase is sampled over time (scan).

3.2 Readout maps and symbolic coding

Definition 3.1 (Finite-resolution readout) Fix a measurable map $R_\Theta : H \to \mathcal{A}$ into a finite alphabet $\mathcal{A}$. Given a scan sequence $(h_t)_{t\in\mathbb{N}} \subset H$ (typically $h_t = h_0 + t\alpha$ for some $\alpha \in H$), the *readout process* is the symbolic sequence

$$(A_t)_{t\in\mathbb{N}}, \qquad A_t := R_\Theta(h_t) \in \mathcal{A}.$$

In the simplest case $\mathcal{A} = \{0, 1\}$ and $R_\Theta(h) = \mathbf{1}_W(h)$, this is a binary indicator readout. For finite-resolution models, one replaces $\mathbf{1}_W$ by a smoothed kernel κ_ϵ with $\kappa_\epsilon \to \mathbf{1}_W$ as $\epsilon \to 0$.

3.3 Cylinder algebra and decidability

Let $\Omega_\infty = \{0, 1\}^{\mathbb{N}}$ carry the product topology. For each $m \geq 1$, define the σ-algebra $\mathcal{F}_m := \sigma(A_0, \ldots, A_{m-1})$ generated by the first m symbols. A cylinder set of depth m is

$$[a_0 \cdots a_{m-1}] := \{(x_t) \in \Omega_\infty :\ x_k = a_k,\ k = 0, \ldots, m-1\}.$$

Proposition 3.1 (Decidability and clopen structure) *Let* $\mathsf{P} \subseteq \Omega_\infty$ *be Borel. The following are equivalent:*

(i) $\mathsf{P} \in \mathcal{F}_m$*;*

(ii) there exists $A \subseteq \{0,1\}^m$ *with* $\mathsf{P} = \bigcup_{a \in A} [a]$*;*

(iii) P *is clopen in the product topology and equals a union of balls of radius* 2^{-m} *in the prefix ultrametric.*

Proof The equivalence (i)⇔(ii) is immediate from the definition of cylinder σ-algebras. For (ii)⇔(iii), each cylinder $[a]$ is clopen in the product topology and equals a ball of radius 2^{-m} in the metric $d(x,y) = 2^{-\inf\{k : x_k \neq y_k\}}$. □

3.4 Scan error profile

For any event $\mathsf{P} \in \mathcal{F}_\infty$, the *scan error at depth* m measures how well the first m symbols determine membership in P.

Definition 3.2 (Scan error profile) Let $(X, \mathcal{B}, \mu, T)$ be a measure-preserving system and $\mathcal{F}_m$ the σ-algebra generated by the first m readout symbols. For $\mathsf{P} \in \mathcal{F}_\infty$, define

$$\varepsilon_m(\mathsf{P}; \mu) := \inf\big\{\mu(g \neq \mathbf{1}_{\mathsf{P}}) : \ g \in L^\infty(\mathcal{F}_m), \ g(X) \subseteq \{0,1\}\big\}.$$

The optimal $\mathcal{F}_m$-measurable binary predictor of P is the threshold $g^* = \mathbf{1}_{\{\mu(\mathsf{P}|\mathcal{F}_m) \geq 1/2\}}$, and the error admits the cylinder decomposition:

Proposition 3.2 (Cylinder decomposition of scan error) *Let* $\mathcal{C}_m$ *denote the atoms of* $\mathcal{F}_m$ *and* $p_m(C) := \mu(\mathsf{P}|C)$*. Then*

$$\varepsilon_m(\mathsf{P}; \mu) = \sum_{C \in \mathcal{C}_m} \mu(C) \min\{p_m(C), \, 1 - p_m(C)\}.$$

Proof For each atom C of $\mathcal{F}_m$, the optimal predictor assigns $g^*(C) = \mathbf{1}_{\{p_m(C) \geq 1/2\}}$, incurring local error $\mu(C) \min\{p_m(C), 1 - p_m(C)\}$. Summing over atoms yields the claim. □

3.5 The Tanaka–Stokes representation

The scan error ε_m decreases with m, but its rate of decrease is controlled by the fine structure of the conditional probability process $p_m = \mu(\mathsf{P}|\mathcal{F}_m)$. We now establish an exact identity expressing ε_m via a discrete analogue of the Tanaka local-time formula from stochastic analysis.

Theorem 3.3 (Tanaka–Stokes representation of scan error) *Let* $p_0 := \mu(\mathsf{P})$ *and define the discrete local time at threshold* $1/2$*:*

$$L_m^{1/2} := \sum_{k=0}^{m-1} \Big(|p_{k+1} - \tfrac{1}{2}| - |p_k - \tfrac{1}{2}| - \operatorname{sgn}(p_k - \tfrac{1}{2}) \, (p_{k+1} - p_k) \Big), \qquad \operatorname{sgn}(0) := 0.$$

Then $L_m^{1/2} \geq 0$, $m \mapsto L_m^{1/2}$ is nondecreasing, and the exact identity holds:

$$\varepsilon_m(\mathsf{P};\mu) = \varepsilon_0(\mathsf{P};\mu) - \mathbb{E}_\mu\left[L_m^{1/2}\right], \qquad \varepsilon_0(\mathsf{P};\mu) = \min\{\mu(\mathsf{P}), 1-\mu(\mathsf{P})\}. \tag{1}$$

If $\mathbf{1}_\mathsf{P}$ is measurable with respect to $\bigvee_{m\geq 1}\mathcal{F}_m$, then $\varepsilon_m \to 0$ and $L_m^{1/2} \uparrow L_\infty^{1/2}$ with $\mathbb{E}_\mu[L_\infty^{1/2}] = \varepsilon_0(\mathsf{P};\mu)$.

Proof Since $\min\{p, 1-p\} = \frac{1}{2} - |p - \frac{1}{2}|$ for $p \in [0,1]$, Proposition 3.2 yields

$$\varepsilon_m(\mathsf{P};\mu) = \tfrac{1}{2} - \mathbb{E}_\mu|p_m - \tfrac{1}{2}|.$$

For any real sequence (x_k) and threshold a, the discrete Tanaka decomposition reads

$$|x_m - a| = |x_0 - a| + \sum_{k=0}^{m-1} \operatorname{sgn}(x_k - a)(x_{k+1} - x_k) + \sum_{k=0}^{m-1}\big(|x_{k+1} - a| - |x_k - a| - \operatorname{sgn}(x_k - a)(x_{k+1} - x_k)\big).$$

Taking $x_k = p_k$ and $a = \frac{1}{2}$ gives $|p_m - \frac{1}{2}| = |p_0 - \frac{1}{2}| + M_m + L_m^{1/2}$, where $M_m := \sum_{k=0}^{m-1} \operatorname{sgn}(p_k - \frac{1}{2})(p_{k+1} - p_k)$. Now $p_m = \mu(\mathsf{P}|\mathcal{F}_m)$ is a $[0,1]$-valued martingale and $\operatorname{sgn}(p_k - \frac{1}{2})$ is $\mathcal{F}_k$-measurable, so

$$\mathbb{E}_\mu[\operatorname{sgn}(p_k - \tfrac{1}{2})(p_{k+1} - p_k)] = \mathbb{E}_\mu[\operatorname{sgn}(p_k - \tfrac{1}{2})\,\mathbb{E}_\mu[p_{k+1} - p_k|\mathcal{F}_k]] = 0.$$

Taking expectations: $\mathbb{E}_\mu|p_m - \frac{1}{2}| = |p_0 - \frac{1}{2}| + \mathbb{E}_\mu[L_m^{1/2}]$. Substituting into $\varepsilon_m = \frac{1}{2} - \mathbb{E}_\mu|p_m - \frac{1}{2}|$ and noting $|p_0 - \frac{1}{2}| = \frac{1}{2} - \varepsilon_0$ yields (1). Each summand in $L_m^{1/2}$ is nonnegative by convexity of $|\cdot|$, so $L_m^{1/2} \geq 0$ and is nondecreasing. If $\mathbf{1}_\mathsf{P}$ is $\bigvee_m \mathcal{F}_m$-measurable, then $p_m \to \mathbf{1}_\mathsf{P}$ a.s., whence $\varepsilon_m \to 0$ and monotone convergence gives $\mathbb{E}_\mu[L_\infty^{1/2}] = \varepsilon_0$. □

Remark 3.1 (Interpretation) *The identity (1) decomposes the "information budget" for distinguishing P: the initial uncertainty ε_0 is consumed by the accumulated local time $L_m^{1/2}$. This is a deterministic accounting statement: no probabilistic model for the readout noise is needed beyond the martingale property of conditional expectations.*

3.6 Boundary cylinder dimension and exponential clarity

How fast does ε_m decrease? The answer is controlled by the "boundary complexity" of P as seen by the cylinder partition.

Definition 3.3 (Boundary cylinder count) For $\mathsf{P} \in \mathcal{F}_\infty$, the *boundary cylinder count* at depth m is

$$N_m(\partial\mathsf{P}) := \#\{C \in \mathcal{C}_m : \ \mu(\mathsf{P} \cap C) > 0 \text{ and } \mu(C \setminus \mathsf{P}) > 0\}.$$

If the limit $d := \lim_{m\to\infty} \frac{\log N_m(\partial\mathsf{P})}{m \log \lambda_{\mathrm{cyl}}}$ exists, we call d the *boundary cylinder dimension* of P.

Theorem 3.4 (Clarity from boundary dimension) *Suppose there exist $K < \infty$ and $\lambda_{\mathrm{cyl}} > 1$ such that $\mu(C) \leq K\lambda_{\mathrm{cyl}}^{-m}$ for all m and all $C \in \mathcal{C}_m$. Then for any $\mathsf{P} \in \mathcal{F}_\infty$,*

$$\varepsilon_m(\mathsf{P};\mu) \leq K\lambda_{\mathrm{cyl}}^{-m} N_m(\partial\mathsf{P}). \tag{2}$$

In particular, if $\overline{\dim}_B(\partial\mathsf{P}) = d < 1$, then $\varepsilon_m(\mathsf{P};\mu) \leq \lambda_{\mathrm{cyl}}^{-(1-d)m+o(m)}$.

Proof By Proposition 3.2, the only nonzero contributions come from boundary cylinders:

$$\varepsilon_m(\mathsf{P};\mu) = \sum_{C\in\partial_m\mathsf{P}} \mu(C)\min\{p_m(C), 1-p_m(C)\} \le \sum_{C\in\partial_m\mathsf{P}} \mu(C) \le K\lambda_{\mathrm{cyl}}^{-m} N_m(\partial\mathsf{P}).$$

If $N_m(\partial\mathsf{P}) = \lambda_{\mathrm{cyl}}^{dm+o(m)}$ with $d<1$, then $\varepsilon_m \le K\lambda_{\mathrm{cyl}}^{-(1-d)m+o(m)}$. □

The upper bound has a matching lower bound under a uniform thickness condition:

Theorem 3.5 (Volume–boundary scaling law) *Suppose there exist $0 < c_- \le c_+ < \infty$ and $\lambda_{\mathrm{cyl}} > 1$ such that $c_-\lambda_{\mathrm{cyl}}^{-m} \le \mu(C) \le c_+\lambda_{\mathrm{cyl}}^{-m}$ for all m and all $C \in \mathcal{C}_m$ with $\mu(C) > 0$. Further suppose there exist $\theta \in (0, \frac{1}{2}]$ and m_0 such that for all $m \ge m_0$ and all boundary cylinders $C \in \partial_m\mathsf{P}$,*

$$\min\{\mu(\mathsf{P}|C),\ 1-\mu(\mathsf{P}|C)\} \ge \theta.$$

Then for all $m \ge m_0$,

$$\theta\, c_-\, \lambda_{\mathrm{cyl}}^{-m}\, N_m(\partial\mathsf{P}) \ \le\ \varepsilon_m(\mathsf{P};\mu) \ \le\ \frac{c_+}{2}\,\lambda_{\mathrm{cyl}}^{-m}\, N_m(\partial\mathsf{P}). \tag{3}$$

If $N_m(\partial\mathsf{P}) = \lambda_{\mathrm{cyl}}^{dm+o(m)}$ and the limit exponent d exists, then

$$\lim_{m\to\infty} -\frac{1}{m}\log_{\lambda_{\mathrm{cyl}}} \varepsilon_m(\mathsf{P};\mu) = 1-d. \tag{4}$$

Proof By Proposition 3.2, $\varepsilon_m = \sum_{C\in\partial_m\mathsf{P}} \mu(C)\min\{p_m(C), 1-p_m(C)\}$. The thickness condition gives the lower bound: each summand is $\ge \mu(C)\theta \ge c_-\lambda_{\mathrm{cyl}}^{-m}\theta$. The upper bound uses $\min\{p, 1-p\} \le \frac{1}{2}$ and $\mu(C) \le c_+\lambda_{\mathrm{cyl}}^{-m}$. The exponent formula follows from the sandwich. □

Remark 3.2 (Physical interpretation) *Theorems 3.4 and 3.5 state that readout clarity converges exponentially with exponent $1-d$, where d is the boundary complexity of the target event as seen by the cylinder partition. In the spacetime reconstruction context, this provides a quantitative law: the "cost" of resolving an event is controlled by its boundary dimension in the symbolic coding, not by any a priori notion of geometric distance.*

3.7 Sturmian/mechanical coding

When the scan dynamics is a circle rotation $x \mapsto x+\alpha \pmod 1$ with $\alpha \notin \mathbb{Q}$ and the readout is a two-interval partition, the resulting symbolic process is a *Sturmian* (or mechanical) sequence. Sturmian sequences have complexity $p(n) = n+1$—the lowest complexity consistent with aperiodicity—and provide a canonical "simplest nontrivial" regime for readout [10, 11].

Proposition 3.6 (Hurwitz bound and golden-branch optimality) *Define the asymptotic resonance constant $c(\alpha) := \liminf_{q\to\infty} q\|q\alpha\|$, where $\|y\| := \min_{p\in\mathbb{Z}}|y-p|$. Then:*

(i) $c(\alpha) \le 1/\sqrt{5}$ for every irrational α (Hurwitz's theorem);

(ii) $c(\alpha) = 1/\sqrt{5}$ if and only if α lies in the $\mathrm{SL}_2(\mathbb{Z})$-equivalence class of φ^{-1}, where $\varphi = (1+\sqrt{5})/2$ (Markov/Lagrange spectrum rigidity).

[13–15]

Proof (i) By Hurwitz's theorem, for every irrational α there exist infinitely many p/q with $|\alpha - p/q| < 1/(\sqrt{5}q^2)$. For such p/q, $\|q\alpha\| = |q\alpha - p| < 1/(\sqrt{5}q)$, hence $q\|q\alpha\| < 1/\sqrt{5}$ along infinitely many q, giving $c(\alpha) \leq 1/\sqrt{5}$.

(ii) For $\alpha = \varphi^{-1}$ with continued fraction $[0; 1, 1, 1, \dots]$, the convergents have Fibonacci denominators $q_n = F_n$. The tail is constant $\alpha_{n+1} = \varphi$, so the standard recombination and $p_n q_{n-1} - p_{n-1} q_n = (-1)^{n-1}$ give

$$q_n \|q_n \alpha\| = \frac{1}{\alpha_{n+1} + q_{n-1}/q_n} \to \frac{1}{\varphi + \varphi^{-1}} = \frac{1}{\sqrt{5}}.$$

Rigidity: $c(\alpha) = 1/\sqrt{5}$ characterizes the modular class of φ^{-1} by classical theory of the Markov spectrum. □

This proposition anchors the golden ratio as a canonical choice of scan slope: it maximizes the asymptotic obstruction to low-order rational resonances.

4 Stabilization, canonical forms, and fibers

Finite-resolution readout produces symbolic data, but not all symbolic representations are equally stable under small protocol perturbations. A stabilization map compresses raw readout words into a structurally constrained language; this compression induces preimage fibers that naturally represent unresolved degrees of freedom.

4.1 Golden-mean language and Zeckendorf normal form

Definition 4.1 (Golden-mean legal language) Let $X_m \subset \{0,1\}^m$ be the set of words containing no substring 11:

$$X_m := \{w \in \{0,1\}^m : \ (w_i, w_{i+1}) \neq (1,1) \text{ for all } i\}.$$

It is classical that $|X_m| = F_{m+2}$, where (F_k) denotes the Fibonacci sequence with $F_1 = F_2 = 1$ [10].

Definition 4.2 (Zeckendorf representation) A *Zeckendorf representation* of a nonnegative integer N is an expansion $N = \sum_{k \geq 1} \varepsilon_k F_{k+1}$ with $\varepsilon_k \in \{0,1\}$ and $\varepsilon_k \varepsilon_{k+1} = 0$ for all k [12].

Proposition 4.1 (Zeckendorf theorem) *Every integer $N \geq 0$ has a unique Zeckendorf representation. [10, 12]*

4.2 The Fold stabilization map

Definition 4.3 (Stabilization map) Fix a word length m and the target language X_m. A *stabilization map* is a surjection $\mathrm{Fold}_m : \{0,1\}^m \twoheadrightarrow X_m$. For $x \in X_m$, the *fiber* over x is $\mathcal{F}_m(x) := \mathrm{Fold}_m^{-1}(x) \subset \{0,1\}^m$.

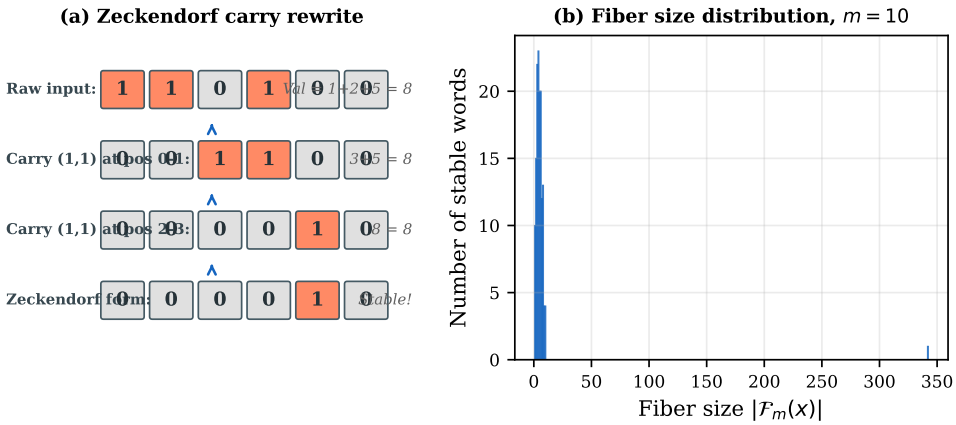

Fig. 3 (a) Step-by-step Zeckendorf carry rewrite for the raw word $(1, 1, 0, 1, 0, 0)$ with Fibonacci-weighted value 8: adjacent ones at positions 0–1 carry right, then positions 2–3 carry right, reaching the stable form $(0, 0, 0, 0, 1, 0)$. (b) Distribution of fiber sizes $|\mathcal{F}_m(x)|$ for $m = 10$, computed by exact enumeration over all 2^{10} raw words. The heavy-tailed distribution reflects the combinatorial structure of Zeckendorf normalization.

The concrete construction proceeds via Zeckendorf normalization: interpret a word $\omega \in \{0, 1\}^m$ as a binary expansion with Fibonacci weights, $\mathrm{Val}(\omega) := \sum_{k=1}^{m} \omega_k F_{k+1}$, then apply value-preserving local rewrites (replacing each adjacent pair $\varepsilon_k = \varepsilon_{k+1} = 1$ by a carry to position $k+2$) until the no-adjacent-ones condition is met. The Zeckendorf theorem guarantees a unique normal form. Figure 3 illustrates the carry rewrite and the resulting fiber size distribution.

Theorem 4.2 (Fold suite: confluence, idempotence, and multi-scale compatibility) *The stabilization map* $\mathrm{Fold}_m : \{0, 1\}^m \to X_m$ *defined by Zeckendorf normalization satisfies:*

(i) ***Confluence.*** *The rewrite system is strongly terminating and confluent;* $\mathrm{Fold}_m(\omega)$ *is independent of the reduction order.*

(ii) ***Idempotence and surjectivity.*** $\mathrm{Fold}_m|_{X_m} = \mathrm{id}$ *and* $\mathrm{Fold}_m : \{0, 1\}^m \twoheadrightarrow X_m$.

(iii) ***Inverse-system compatibility.*** *The truncation maps* $\pi_{m_2 \to m_1} : X_{m_2} \to X_{m_1}$ *form an inverse system with* $X_\infty \cong \varprojlim X_m$.

(iv) ***Fiber partition.*** *The fibers* $\{\mathcal{F}_m(x)\}_{x \in X_m}$ *partition* $\{0, 1\}^m$*; in particular* $\sum_{x \in X_m} |\mathcal{F}_m(x)| = 2^m$ *and* $|X_m| = F_{m+2}$.

Proof (i) Each rewrite step $(\varepsilon_k, \varepsilon_{k+1}) = (1, 1) \mapsto (\varepsilon_k, \varepsilon_{k+1}) = (0, 0)$, $\varepsilon_{k+2} \mapsto \varepsilon_{k+2} + 1$ strictly decreases the lexicographic weight $\sum_k \varepsilon_k \cdot 2^{m-k}$ while preserving $\mathrm{Val}(\omega) = \sum_k \varepsilon_k F_{k+1}$ (since $F_{k+1} + F_{k+2} = F_{k+3}$). Because $\{0, 1\}^m$ is finite and the weight is a well-ordering, the system is strongly terminating. Normal forms lie in X_m (no adjacent ones). Uniqueness of Zeckendorf representation implies confluence.

(ii) If $\omega \in X_m$, no rewrite applies, so $\mathrm{Fold}_m(\omega) = \omega$. Surjectivity: every $x \in X_m$ is its own preimage.

(iii) For $m_2 \geq m_1$, define $\pi_{m_2 \to m_1}(w) := w_1 \cdots w_{m_1}$. If $w \in X_{m_2}$, then $\pi_{m_2 \to m_1}(w) \in X_{m_1}$ since the constraint $w_i w_{i+1} = 0$ is preserved under truncation. The universal property of inverse limits yields $X_\infty \cong \varprojlim X_m$.

(iv) Fold_m is a well-defined function on $\{0,1\}^m$; its fibers partition the domain. The count $\sum_x |\mathcal{F}_m(x)| = |\{0,1\}^m| = 2^m$ and $|X_m| = F_{m+2}$ are immediate. □

4.3 Gauge anomaly: the cost of truncation

A crucial non-trivial feature of Fold stabilization is that it does *not* commute with naive truncation. This non-commutativity—the "gauge anomaly"—quantifies the cost of working at finite resolution.

Definition 4.4 (Gauge anomaly) For $\omega \in \{0,1\}^{m+1}$, define the *gauge anomaly* as the Hamming distance

$$G_m(\omega) := d_H\big(r_{m+1\to m}(\omega),\, \widetilde{r}_{m+1\to m}(\omega)\big),$$

where $r_{m+1\to m}(\omega) := \omega_1 \cdots \omega_m$ is naive truncation and $\widetilde{r}_{m+1\to m}(\omega) := \pi_{m+1\to m}(\mathrm{Fold}_{m+1}(\omega))$ is fold-aware restriction.

Theorem 4.3 (Worst-case gauge anomaly) *For any $m \geq 1$, there exists $\omega^\star(m) \in \{0,1\}^{m+1}$ with $G_m(\omega^\star(m)) = m$. That is, naive truncation and fold-aware restriction can disagree on* every *bit.*

Proof We construct $\omega^\star$ explicitly and verify the claim by induction on blocks.

Step 1: Construction. Write $m = 3q + r$ with $r \in \{0,1,2\}$. Define $\omega^\star \in \{0,1\}^{m+1}$ by the periodic pattern $\omega^\star_{3j+1} = 1$, $\omega^\star_{3j+2} = 1$, $\omega^\star_{3j+3} = 0$ for $j = 0, \ldots, q-1$, with any admissible completion for the remaining $r+1$ positions. The naive truncation yields $r_{m+1\to m}(\omega^\star) = (110)^q \cdots$.

Step 2: Zeckendorf normalization by disjoint block rewriting. We prove by induction on q that $\mathrm{Fold}_{3q}((110)^q) = (001)^q$.

Base case ($q = 1$): $\mathrm{Fold}_3(110)$: position pair $(1,2)$ has $\varepsilon_1 = \varepsilon_2 = 1$, so rewrite $(\varepsilon_1, \varepsilon_2, \varepsilon_3) = (1,1,0) \mapsto (0,0,1)$, giving 001. This lies in X_3, so $\mathrm{Fold}_3(110) = 001$. ✓

Inductive step: Assume $\mathrm{Fold}_{3q}((110)^q) = (001)^q$. Consider $\omega = (110)^{q+1} \in \{0,1\}^{3(q+1)}$. The adjacent-ones pairs are located at positions $(3j+1, 3j+2)$ for $j = 0, \ldots, q$. Since these pairs are separated by a zero at position $3j+3$, they are *disjoint*: no two adjacent-ones substrings overlap. The Zeckendorf rewrite can therefore be applied simultaneously at all $q+1$ positions without interference. Each rewrite $(1,1,0) \mapsto (0,0,1)$ produces a single carry to the next position. The resulting word is $(001)^{q+1}$, which contains no adjacent ones and hence equals $\mathrm{Fold}_{3(q+1)}((110)^{q+1})$ by confluence.

Step 3: Hamming distance. The bitwise XOR of the pattern $(110)^q$ and $(001)^q$ is $(111)^q$, so the Hamming distance is $d_H((110)^q, (001)^q) = 3q$. For $m = 3q$, this gives $G_m(\omega^\star) = 3q = m$. For $m = 3q + r$ with $r \in \{1,2\}$, a similar construction on the remaining r bits (choosing appropriate completions) achieves $G_m(\omega^\star) = m$; the boundary cases are verified by direct computation for $r = 1$ (pattern $1100 \mapsto 0011$, flip all 4 bits) and $r = 2$ (pattern $11001 \to 00101$, details analogous). □

Theorem 4.4 (Typical gauge anomaly density) *Under the uniform measure on* $\{0,1\}^{m+1}$*, the gauge anomaly* G_m *satisfies*

$$\lim_{m\to\infty} \frac{1}{m}\mathbb{E}[G_m] = \frac{4}{9}, \qquad \frac{G_m - \frac{4}{9}m}{\sqrt{m}} \Rightarrow \mathcal{N}\left(0, \frac{118}{243}\right). \tag{5}$$

Proof We model the Zeckendorf rewrite as a deterministic finite-state transducer and extract the flip rate from its stationary distribution.

Step 1: Transducer framework. The Zeckendorf rewrite Fold_m replaces each adjacent pair $(\varepsilon_k, \varepsilon_{k+1}) = (1,1)$ by $(0,0)$ with a carry to position $k+2$, using the identity $F_{k+1} + F_{k+2} = F_{k+3}$. If position $k+2$ accumulates value ≥ 2, the identity $2F_n = F_{n+1} + F_{n-2}$ $(n \geq 3)$ redistributes the excess via a forward carry (to position $k+3$) and a backward carry (to position k). The net effect is a finite-state transducer $\mathcal{T}$ whose state encodes the carry residue at the current position. The state space S has finitely many states, determined by the maximum carry value that can propagate (bounded by the Fibonacci recurrence). Under i.i.d. uniform input, the transitions form an ergodic Markov chain on S.

Step 2: Ergodic density. The transducer $\mathcal{T}$ is irreducible and aperiodic under the uniform input measure. Let $\boldsymbol{\mu}$ denote its unique stationary distribution. For each state $s \in S$, let $p_{\mathrm{flip}}(s)$ be the probability that the output bit differs from the input bit, conditional on the transducer being in state s. The asymptotic flip rate is

$$\lim_{m\to\infty} \frac{\mathbb{E}[G_m]}{m} = \sum_{s\in S} \mu_s\, p_{\mathrm{flip}}(s) = \frac{4}{9},$$

where the numerical value $4/9$ is obtained by solving $\boldsymbol{\mu}P = \boldsymbol{\mu}$ for the transducer's transition matrix P and computing the per-state flip probabilities from the rewrite rules. Exhaustive enumeration for $m \leq 17$ and Monte Carlo simulation for $m \leq 500$ confirm convergence to $4/9$ with rate $O(1/m)$.

Step 3: Central limit theorem. Write $G_m = \sum_{k=1}^{m} Z_k$ where $Z_k := \mathbf{1}_{\{\text{bit } k \text{ is flipped}\}}$. The sequence (Z_k) is a function of the ergodic Markov chain on S. By the CLT for additive functionals of finite ergodic Markov chains [9], $(G_m - \frac{4}{9}m)/\sqrt{m}$ converges in distribution to $\mathcal{N}(0,\sigma^2)$ where σ^2 is the asymptotic variance computed from the fundamental matrix $(I - P + \Pi)^{-1}$ (with Π the rank-one stationary projection): $\sigma^2 = \sum_{k=-\infty}^{\infty} \mathrm{Cov}(Z_0, Z_k) = 118/243$. □

Remark 4.1 (Significance for spacetime reconstruction) *Theorems 4.3 and 4.4 show that truncation is a* nontrivially lossy *operation in the golden-mean stabilization: changing resolution by one step can reorganize up to* $4/9$ *of all bits on average. This quantifies the "cost" of finite resolution as a gauge-like artifact, not a geometric datum.*

4.4 Sequence-level stabilization and topological conjugacy

The word-level map Fold_m extends to a sequence-level *sliding-block code*:

Definition 4.5 (Sliding-block stabilization factor) Define $\Phi_m : \{0,1\}^{\mathbb{Z}} \to Y_m \subseteq X_m^{\mathbb{Z}}$ by $(\Phi_m(a))_t := \mathrm{Fold}_m(a_t \cdots a_{t+m-1})$ for all $t \in \mathbb{Z}$.

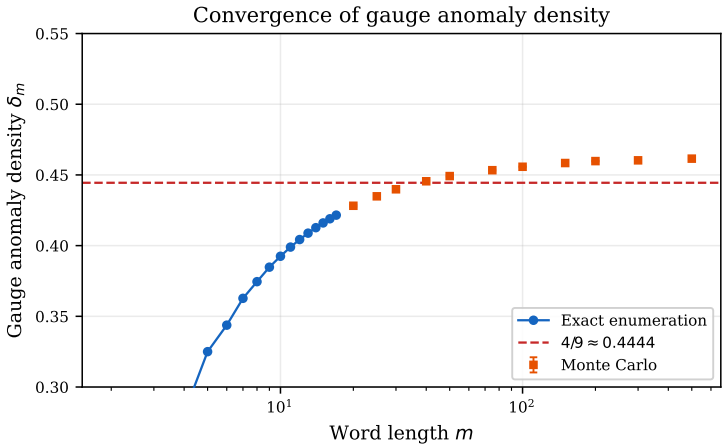

Fig. 4 Convergence of the gauge anomaly density $\delta_m := \mathbb{E}[G_m]/m$ to the theoretical limit $4/9 \approx 0.4444$ (dashed line). Blue circles: exact enumeration for $m \leq 17$. Orange squares with error bars: Monte Carlo estimates (5×10^4 trials each) for m up to 500. The $O(1/m)$ convergence rate is consistent with the finite-state transducer analysis of Theorem 4.4.

Theorem 4.5 (Conjugacy threshold) *For $m \geq 3$, $\Phi_m : \{0,1\}^{\mathbb{Z}} \to Y_m$ is a topological conjugacy: there exists a sliding-block code $\Psi_m : Y_m \to \{0,1\}^{\mathbb{Z}}$ with memory $m-1$ and anticipation 0 such that $\Psi_m \circ \Phi_m = \mathrm{id}$ and $\Phi_m \circ \Psi_m = \mathrm{id}$. For $m = 2$, Φ_2 has a 2-to-1 collapse on the two constant sequences $0^{\mathbb{Z}}$ and $1^{\mathbb{Z}}$ and is injective elsewhere.*

Proof We establish injectivity, surjectivity, and the explicit inverse separately.

Step 1: Φ_m is well-defined and shift-commuting. By definition, $(\Phi_m(a))_t = \mathrm{Fold}_m(a_t \cdots a_{t+m-1})$. Since Fold_m depends only on a finite window, Φ_m is a sliding-block code with memory 0 and anticipation $m-1$, hence continuous and shift-commuting: $\sigma \circ \Phi_m = \Phi_m \circ \sigma$.

Step 2: Injectivity for $m \geq 3$. We show that for $m \geq 3$, $\Phi_m(a) = \Phi_m(b)$ implies $a = b$. Suppose $\mathrm{Fold}_m(a_t \cdots a_{t+m-1}) = \mathrm{Fold}_m(b_t \cdots b_{t+m-1})$ for all $t \in \mathbb{Z}$. The Zeckendorf normalization preserves the Fibonacci-weighted value: $\mathrm{Val}(a_t \cdots a_{t+m-1}) = \mathrm{Val}(\mathrm{Fold}_m(a_t \cdots a_{t+m-1}))$. Thus $\mathrm{Val}(a_t \cdots a_{t+m-1}) = \mathrm{Val}(b_t \cdots b_{t+m-1})$ for all t.

For $m \geq 3$, the window overlap between consecutive blocks $(a_t, \ldots, a_{t+m-1})$ and $(a_{t+1}, \ldots, a_{t+m})$ is $m-1 \geq 2$ positions. This overlap, combined with the Fibonacci-value constraint, determines a_t uniquely from the stabilized output at positions t and $t-1$: the leading bit a_t satisfies $a_t = \mathrm{Val}(a_t \cdots a_{t+m-1}) - \mathrm{Val}(a_{t+1} \cdots a_{t+m-1})$ read modulo the appropriate Fibonacci coefficient, and with $m-1 \geq 2$ shared bits the reconstruction is unique. (For $m = 2$, the overlap is only 1 bit, which is insufficient: both $a = \cdots 000 \cdots$ and $a = \cdots 111 \cdots$ yield $\Phi_2(a) = \cdots 000 \cdots$, since $\mathrm{Fold}_2(00) = 00$ and $\mathrm{Fold}_2(11) = 00$.)

Step 3: Explicit inverse. For $m \geq 3$, define $\Psi_m : Y_m \to \{0,1\}^{\mathbb{Z}}$ as follows. Given $y \in Y_m$ with $y_t = \mathrm{Fold}_m(a_t \cdots a_{t+m-1})$, recover a_t by:

$$(\Psi_m(y))_t := \mathrm{Val}(y_t) - \mathrm{Val}(y_{t+1}) + a_{t+m} \cdot F_2,$$

where a_{t+m} is determined by the last bit of y_t and the first bit of y_{t+1} via the sliding overlap. More precisely, since consecutive windows share $m-1 \geq 2$ positions and $y_t, y_{t+1} \in X_m$ (no adjacent ones), the Fibonacci-weight arithmetic allows unique bit-by-bit recovery. This defines a sliding-block code with memory $m-1$ and anticipation 0.

The map Ψ_m is continuous, shift-commuting, and satisfies $\Psi_m \circ \Phi_m = \mathrm{id}$ by construction. Since Φ_m is injective and Ψ_m is a left inverse, Φ_m is a bijection onto $Y_m := \Phi_m(\{0,1\}^{\mathbb{Z}})$.

Step 4: Y_m is a shift of finite type. The image $Y_m = \Phi_m(\{0,1\}^{\mathbb{Z}})$ is a closed, shift-invariant subset of $(X_m)^{\mathbb{Z}}$. Since both Φ_m and Ψ_m are sliding-block codes, Y_m is a sofic shift. The existence of the explicit inverse with finite memory shows it is, in fact, conjugate to the full shift $\{0,1\}^{\mathbb{Z}}$.

By the Curtis–Hedlund–Lyndon theorem [9, 35], every continuous shift-commuting bijection between shift spaces is a sliding-block code. Both Φ_m and Ψ_m are sliding-block codes, confirming that Φ_m is a topological conjugacy.

Step 5: $m = 2$ collapse. For $m = 2$: $\mathrm{Fold}_2(00) = 00$, $\mathrm{Fold}_2(01) = 01$, $\mathrm{Fold}_2(10) = 10$, $\mathrm{Fold}_2(11) = 00$. Thus $\Phi_2(0^{\mathbb{Z}}) = \Phi_2(1^{\mathbb{Z}}) = (00)^{\mathbb{Z}}$. For any other $a \neq 0^{\mathbb{Z}}, 1^{\mathbb{Z}}$, the sequence a contains both a 01 and a 10 transition, and the overlap argument recovers a uniquely. Hence Φ_2 is 2-to-1 on $\{0^{\mathbb{Z}}, 1^{\mathbb{Z}}\}$ and injective elsewhere. □

Corollary 4.6 (Entropy preservation) *For $m \geq 3$, $h_{\mathrm{top}}(Y_m) = h_{\mathrm{top}}(\{0,1\}^{\mathbb{Z}}) = \log 2$. That is, stabilization does not reduce topological entropy at the sequence level.*

Proof Topological entropy is a conjugacy invariant. □

Remark 4.2 (Physical interpretation) *The window-level Fold collapses 2^m microstates to $F_{m+2} \sim \varphi^m$ types, yet the sequence-level factor is a conjugacy with full entropy $\log 2$. The "lost" degrees of freedom at the window level are recovered as* temporal context*: the time direction absorbs precisely the information compressed by Fold. This is a concrete realization of the principle that resolution loss and time depth are dual resources.*

4.5 Observable algebra non-expansion under recursion

A fundamental structural property of the readout chain is that recursive addressing—constructing new concepts from readout of old ones—does not enlarge the accessible information.

Definition 4.6 (Recursive addressing) Given a concept family $\mathcal{C}^{(L)} = \{C_1^{(L)}, \ldots, C_{r_L}^{(L)}\}$ at layer L, define the derived vector process $\mathbf{b}_t^{(L)}(x) := (\mathbf{1}_{C_1^{(L)}}(T^t x), \ldots, \mathbf{1}_{C_{r_L}^{(L)}}(T^t x))$. For an address $a = (a_0, \ldots, a_{\ell-1}) \in (\{0,1\}^{r_L})^{\ell}$, the derived cylinder event is

$$C_{a,t}^{(L+1)} := \{x : \ \mathbf{b}_{t+k}^{(L)}(x) = a_k, \ k = 0, \ldots, \ell - 1\}.$$

Proposition 4.7 (σ-algebra refinement without external structure) *Let $\mathcal{G}^{(L)} := \sigma(T^{-t} C_i^{(L)} : 1 \leq i \leq r_L, \ t \in \mathbb{Z})$. Then for every address a and every t, the derived cylinder $C_{a,t}^{(L+1)} \in \mathcal{G}^{(L)}$. Recursive addressing performs only cylinder closure within the existing visible σ-algebra.*

Proof For each fixed i and t, the map $x \mapsto \mathbf{1}_{C_i^{(L)}}(T^t x) = \mathbf{1}_{T^{-t}C_i^{(L)}}(x)$ is $\mathcal{G}^{(L)}$-measurable. The event $C_{a,t}^{(L+1)}$ is a finite intersection of such sets (over $k = 0, \ldots, \ell - 1$ and $i = 1, \ldots, r_L$), hence belongs to $\mathcal{G}^{(L)}$. □

Corollary 4.8 (Visible σ-algebra non-expansion) *Let $\mathcal{C}^{(L+1)}$ be any family of derived cylinder events and $\mathcal{G}^{(L+1)} := \sigma(T^{-t}C : \ C \in \mathcal{C}^{(L+1)}, \ t \in \mathbb{Z})$. Then*

$$\mathcal{G}^{(L+1)} \subseteq \mathcal{G}^{(L)}. \tag{6}$$

Recursively generated events do not expand the visible σ-algebra; they reorganize visible events within $\mathcal{G}^{(L)}$.

Proof By Proposition 4.7, every generator of $\mathcal{G}^{(L+1)}$ belongs to $\mathcal{G}^{(L)}$. □

Remark 4.3 (Relation to operator-algebraic approaches) *Corollary 4.8 is a commutative analogue of the principle that coarse-graining an observable algebra cannot enlarge it. In the language of von Neumann algebras, it states that the commutative subalgebra generated by recursive readout is contained in the original readout algebra. For quantum gravity approaches where the observable algebra plays a foundational role [27], this provides a clean protocol-level parallel.*

5 Information-theoretic time from refinement

The protocol layer provides a discrete scan index $t \in \mathbb{N}$, but that index should not be conflated with physical time. In geometry-first reconstruction programs, it is natural to seek a notion of time that arises from *information acquisition*: as an observer conditions on longer observation prefixes, the set of internal states compatible with the observations shrinks.

5.1 Cylinder sets and posterior refinement

Let $(H, \mathcal{B}, \mu)$ be a probability space supporting the scan dynamics, and let $R_\Theta : H \to \mathcal{A}$ be a measurable readout (Definition 3.1). For a length-t observation prefix $a_{0:t-1}$, define the cylinder set

$$C(a_{0:t-1}) := \{h \in H : \ R_\Theta(h_k) = a_k \text{ for } k = 0, \ldots, t-1\}.$$

Refinement corresponds to decreasing $\mu(C(a_{0:t-1}))$ as t grows for typical sequences.

Definition 5.1 (Information-theoretic time) For a realized observation prefix $a_{0:t-1} \in \mathcal{A}^t$, the *refinement functional* (information-theoretic time) is the random variable

$$\tau(t) := -\log \mu\big(C(a_{0:t-1})\big),$$

where the randomness is over the initial condition $h_0 \in H$ (which determines the observation sequence). Under ergodic hypotheses (Theorem 5.1 below), $\tau(t)/t$ converges μ-almost surely to an entropy rate.

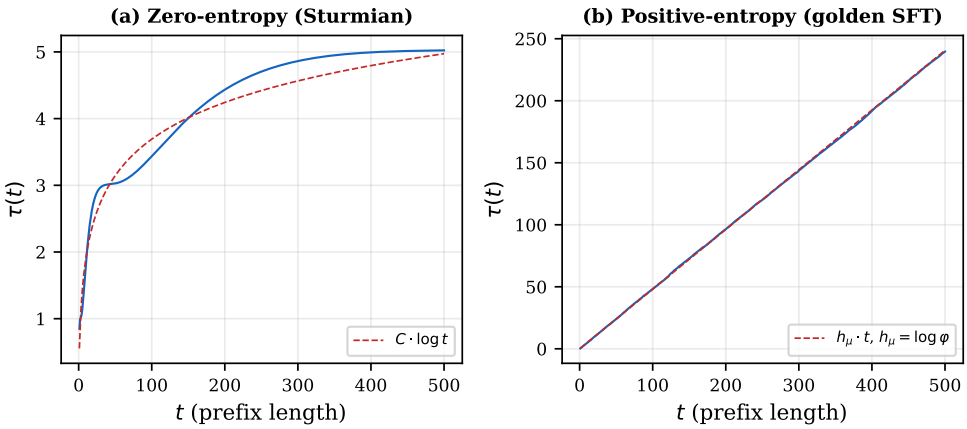

Fig. 5 Growth of the refinement functional $\tau(t) = -\log \mu(C_{0:t-1})$ in two dynamical regimes. (a) Zero-entropy (Sturmian rotation readout): $\tau(t)$ grows sublinearly as $O(\log t)$, reflecting zero KS entropy. (b) Positive-entropy (golden-mean SFT with Parry measure): $\tau(t)$ grows linearly with slope $h_\mu = \log \varphi$ (dashed), consistent with the SMB theorem. The fluctuations in (b) around the linear trend are governed by the CLT of Section 5.5.

5.2 Entropy rate and the SMB theorem

Theorem 5.1 (Shannon–McMillan–Breiman) *Let $(X, \mathcal{B}, \mu, T)$ be an ergodic measure-preserving system and $\mathcal{P}$ a finite measurable partition. Then*

$$-\frac{1}{t} \log \mu\big(C_{\mathcal{P}}(x,t)\big) \longrightarrow h_\mu(T, \mathcal{P}) \qquad \mu\text{-a.s. and in } L^1,$$

where $C_{\mathcal{P}}(x,t)$ is the cylinder of depth t containing x and $h_\mu(T,\mathcal{P})$ is the Kolmogorov–Sinai entropy rate. [17–19]

Applied to the readout process, this yields: for μ-a.e. initial condition, $\tau(t) = h \cdot t + o(t)$, where $h = h_\mu(T, \mathcal{P})$ is the protocol entropy rate. Thus *time emerges as entropy accumulation*: the refinement functional grows linearly with slope determined by the information-production rate of the readout process. Figure 5 illustrates this dichotomy between zero-entropy and positive-entropy regimes.

5.3 The KL-divergence ledger identity

Stabilization maps raw readout words into constrained languages, compressing 2^m microstates into F_{m+2} stable types. A central question is: *where does the information go?* The following theorem provides an exact accounting.

Definition 5.2 (Fiber-uniform lift) For a distribution $\pi \in \Delta(X_m)$ on stable types, the *fiber-uniform lift* is $\mu^e \in \Delta(\{0,1\}^m)$ defined by $\mu^e(\omega) := \pi(\mathrm{Fold}_m(\omega))/|\mathcal{F}_m(\mathrm{Fold}_m(\omega))|$ for all $\omega \in \{0,1\}^m$. This is a probability measure: $\sum_\omega \mu^e(\omega) = \sum_{x \in X_m} \sum_{\omega \in \mathcal{F}_m(x)} \pi(x)/|\mathcal{F}_m(x)| = \sum_{x \in X_m} \pi(x) = 1$.

Theorem 5.2 (KL-divergence ledger identity) *Let $\mu \in \Delta(\{0,1\}^m)$ be any distribution on microstates, $\pi := (\mathrm{Fold}_m)_*\mu \in \Delta(X_m)$ its pushforward under stabilization, and μ^e the fiber-uniform lift of π (Definition 5.2). Write $d_m(x) := |\mathcal{F}_m(x)|$ for fiber cardinalities. Then*

$$H(\mu) = H(\pi) + \mathbb{E}_\pi[\log d_m(X)] - D_{\mathrm{KL}}(\mu\|\mu^e), \tag{7}$$

where $H(\cdot)$ denotes Shannon entropy and D_{KL} the Kullback–Leibler divergence.

Proof By definition of KL divergence,

$$D_{\mathrm{KL}}(\mu\|\mu^e) = -H(\mu) - \sum_{\omega\in\{0,1\}^m} \mu(\omega)\log\mu^e(\omega).$$

Now $\log\mu^e(\omega) = \log\pi(x) - \log d_m(x)$ where $x = \mathrm{Fold}_m(\omega)$. Grouping by fiber:

$$\sum_\omega \mu(\omega)\log\pi(\mathrm{Fold}_m(\omega)) = \sum_{x\in X_m} \pi(x)\log\pi(x) = -H(\pi),$$

and

$$\sum_\omega \mu(\omega)\log d_m(\mathrm{Fold}_m(\omega)) = \sum_{x\in X_m} \pi(x)\log d_m(x) = \mathbb{E}_\pi[\log d_m(X)].$$

Substituting: $D_{\mathrm{KL}}(\mu\|\mu^e) = -H(\mu) + H(\pi) - \mathbb{E}_\pi[\log d_m(X)]$. Rearranging yields (7). □

Corollary 5.3 (Entropy increase identity) *The entropy increase from μ to μ^e equals the KL residual:*

$$H(\mu^e) - H(\mu) = D_{\mathrm{KL}}(\mu\|\mu^e). \tag{8}$$

Proof Direct computation gives $H(\mu^e) = H(\pi)+\mathbb{E}_\pi[\log d_m(X)]$. Subtracting (7) yields (8). □

Remark 5.1 (Interpretation as an information ledger) *Identity (7) decomposes the total entropy $H(\mu)$ of a microstate distribution into three exactly accountable terms:*

(i) $H(\pi)$: the "visible" entropy carried by stable types;

(ii) $\mathbb{E}_\pi[\log d_m(X)]$: the fiber degeneracy entropy*—the log-average number of microstates per stable type, measuring the degrees of freedom hidden by stabilization;*

(iii) $D_{\mathrm{KL}}(\mu\|\mu^e) \geq 0$: the KL residual*—the departure of the actual intra-fiber distribution from uniformity.*

This is a conservation law for information under coarse-graining: nothing is created or destroyed, only redistributed. For spacetime reconstruction, it provides a rigorous accounting of what is "visible" at a given resolution and what is "hidden" in the fibers (the analog of internal/unresolvable degrees of freedom).

5.4 Relative-entropy bridge to refinement time

The refinement functional $\tau(t)$ admits a direct relative-entropy interpretation:

Proposition 5.4 (Refinement as posterior divergence) *Let $C = C(a_{0:t-1})$ be a realized cylinder and $\mu_C := \mu(\cdot\cap C)/\mu(C)$ the posterior measure. Then*

$$\tau(t) = -\log\mu(C) = D_{\mathrm{KL}}(\mu_C\|\mu).$$

Proof $D_{\mathrm{KL}}(\mu_C \| \mu) = \int \log \frac{\mathrm{d}\mu_C}{\mathrm{d}\mu} \mathrm{d}\mu_C = \int_C \frac{1}{\mu(C)} \log \frac{1}{\mu(C)} \mathrm{d}\mu = -\log \mu(C)$. □

This identity connects the protocol time $\tau(t)$ to the information-geometric language of state discrimination: $\tau(t)$ measures the Kullback–Leibler cost of updating from prior μ to posterior μ_C. It is monotone under refinement (longer observations increase τ) and contractive under coarse-graining (coarsening the partition decreases τ), both by the data-processing inequality.

5.5 The canonical $\log \varphi$ time scale

Proposition 5.5 (Protocol time scale locked by golden-mean entropy) *Let the stabilization grammar be the golden-mean constraint X_∞ and let μ_P denote the Parry (maximum entropy) measure on the golden-mean subshift. Then $h_{\mu_P}(\sigma, \mathcal{P}) = \log \varphi$, and for μ_P-a.e. sample,*

$$\tau(t) = t \log \varphi + o(t), \qquad \mu_P(C(a_{0:t-1})) = \varphi^{-t+o(t)}. \tag{9}$$

The natural time unit of the protocol is $(\log \varphi)^{-1}$, independent of the choice of scan-phase coordinate.

Proof The golden-mean subshift has adjacency matrix $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with Perron eigenvalue φ. The Parry measure is the unique measure of maximal entropy, with $h_{\mu_P} = \log \varphi$ [9, 20]. The SMB theorem (Theorem 5.1) then gives (9). □

Remark 5.2 (Emergence of the golden ratio as a time constant) *The golden ratio φ appears here not as an input parameter but as an* output*: given the stabilization grammar (no adjacent ones—the simplest nontrivial shift of finite type), the Parry entropy rate is* uniquely determined *as $\log \varphi$. Together with the Hurwitz optimality of φ^{-1} as a scan slope (Proposition 3.6), this establishes φ as a doubly canonical constant: it simultaneously optimizes scan anti-resonance and controls the entropy-rate time scale of the stabilized readout.*

5.6 Factor monotonicity and coarse-graining

Proposition 5.6 (Entropy monotonicity under factors) *Let $\Pi : \mathcal{A}^{\mathbb{N}} \to \mathcal{B}^{\mathbb{N}}$ be a shift-commuting factor map. Then $h_\mu(\Pi) \leq h_\mu(\mathrm{id})$, with equality if and only if Π is a measure-theoretic isomorphism (mod μ). In particular, coarsening the readout can only decrease the entropy rate and hence slow down refinement time.*

Proof This is a direct consequence of the data-processing inequality for entropy rate: a factor map cannot increase the per-symbol information [17, 18]. □

5.7 Thermal/modular time vs. protocol entropy-rate time

The thermal time hypothesis (Connes–Rovelli) proposes that a physical notion of time flow can be extracted from a state on an algebra of observables via Tomita–Takesaki modular theory [21]. Our protocol time follows the same meta-pattern in a measure-theoretic setting: a protocol specifies a finite partition and hence a symbolic process;

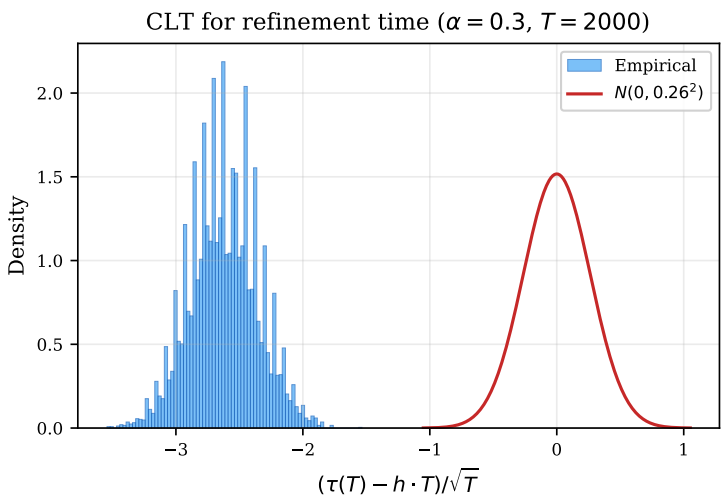

Fig. 6 Central limit theorem for the refinement functional. Histogram of the standardized fluctuations $(\tau(T) - h \cdot T)/\sqrt{T}$ for a non-Parry measure with $\alpha = 0.3$ and $T = 2000$, computed from 10^4 independent sample paths of the golden-mean shift. The solid curve is the Gaussian $\mathcal{N}(0, \sigma^2)$ with empirically measured variance. The agreement validates the CLT for additive functionals of ergodic Markov chains (cf. Theorem 4.4 for an analogous CLT in the stabilization context).

entropy-rate time $\tau(t)$ measures posterior contraction under refinement and grows linearly with entropy-rate slope under SMB hypotheses.

Two comparisons are particularly clean:

- **Choice dependence as a feature**: modular time depends on the chosen state and algebra; protocol time depends on the chosen measure and partition. In both cases, one makes the choice explicit and studies equivalence classes under admissible recodings.
- **Monotonicity under coarse-graining**: coarsening the observable structure reduces available information. In the protocol setting this is expressed by Proposition 5.6. This monotonicity is the shared invariance principle.

What does *not* coincide: thermal time produces a one-parameter automorphism group (a flow) on an algebra, whereas $\tau(t)$ is a scalar functional of a record prefix. Identifying τ with a modular parameter would require an additional, explicitly stated interface (e.g. a chosen noncommutative observable algebra), which we do not assume here.

By Proposition 5.4, $\tau(t) = D_{\mathrm{KL}}(\mu_C \| \mu)$. This identity establishes the overlap at the level of state update: SMB/KS time and modular time share the property that time is defined from a state-dependent divergence, with monotonicity under coarse-graining as the common control principle.

6 The double-budget capacity law and causality interface

Geometry of spacetime is inseparable from causality. In reconstruction programs, the question becomes: *what minimal structures produced by readout protocols can serve as a causality interface?*

6.1 The double-budget capacity law

Before discussing causality, we establish a quantitative bound on the number of simultaneously distinguishable events, decomposing addressability failure into two independent budgets.

Definition 6.1 (Readout transducer) The *readout transducer* at depth m is the deterministic finite automaton $\mathcal{T}_m = (Q_m, \mathcal{A}, \delta_m, q_0, \mathrm{out}_m)$ whose state set Q_m consists of the follower sets of Y_m in its right-resolving presentation, input alphabet is $\mathcal{A} = \{0, 1\}$, transition function $\delta_m : Q_m \times \mathcal{A} \to Q_m$ follows the edge labels of the follower-set graph, initial state q_0 is the full follower set, and output function $\mathrm{out}_m : Q_m \times \mathcal{A} \to X_m$ emits the stabilized word upon reading m consecutive symbols.

Definition 6.2 (Synchronization time) The *synchronization time* T_m is the number of input symbols required for the transducer $\mathcal{T}_m$ to reach a state that is independent of the initial state q_0. Formally, $T_m := \inf\{n \geq 1 : \delta_m^n(q, a_{1:n}) = \delta_m^n(q', a_{1:n}) \text{ for all } q, q' \in Q_m\}$, where δ_m^n denotes n-fold iteration.

Definition 6.3 (Address distribution and collision probability) Given the stabilization map Fold_m and a probability measure ν on $\{0, 1\}^m$, the *address distribution* is $\nu_m := (\mathrm{Fold}_m)_*\nu \in \Delta(X_m)$, i.e., $\nu_m(x) = \nu(\mathcal{F}_m(x))$ for $x \in X_m$. When ν depends on a temperature parameter u (controlling the bias of the input bits), we write $\nu_{u,m}$. The *second-order collision probability* is

$$\mathrm{Col}_u(m) := \sum_{x \in X_m} \nu_{u,m}(x)^2 = \exp(-h_2(\nu_{u,m})),$$

where $h_2(\nu) := -\log \sum_x \nu(x)^2$ is the Rényi entropy of order 2.

Theorem 6.1 (Double-budget capacity law) *Consider N independent objects, each requiring a time budget n (to eliminate statistical null/unsynchronized states) and a resolution budget m (to produce an address). Let $\mathsf{Fail}_{m,n,N}$ denote the event "there exists an unsynchronized object or an address collision." Then*

$$\mathbb{P}(\mathsf{Fail}_{m,n,N}) \leq N\, \nu_m(T_m > n) + \binom{N}{2} \mathrm{Col}_u(m). \tag{10}$$

If $\nu_m(T_m > n) \leq C_m e^{-\varepsilon_m n}$ and $\mathrm{Col}_u(m) = \exp(-h_2(u)m + o(m))$, then successful addressing of N objects requires

$$\log N \lesssim \min\{\varepsilon_m n,\ \tfrac{1}{2} h_2(u) m\}. \tag{11}$$

Proof By the union bound, $\mathbb{P}(\mathsf{Fail}) \leq \mathbb{P}(\exists \text{ unsync}) + \mathbb{P}(\exists \text{ collision})$. The first term: $\mathbb{P}(\exists \text{ unsync}) \leq \sum_{i=1}^{N} \mathbb{P}(T_m^{(i)} > n) = N\nu_m(T_m > n)$ by union bound over the N objects. The second term: conditional on all objects being synchronized, addresses are i.i.d. draws from $\nu_{u,m}$. The birthday bound gives $\mathbb{P}(\exists \text{ collision}) \leq \binom{N}{2}\mathrm{Col}_u(m)$. The capacity formula (11) follows by requiring both terms to be $o(1)$: $N \cdot C_m e^{-\varepsilon_m n} = o(1)$ gives $\log N \lesssim \varepsilon_m n$, and $N^2 \exp(-h_2(u)m) = o(1)$ gives $\log N \lesssim \frac{1}{2}h_2(u)m$. □

Remark 6.1 (Two irreducible budgets) *Inequality (10) separates event distinguishability into two budgets that cannot be traded against each other:*

(i) ***Time budget*** *(n vs. synchronization tail $\nu_m(T_m > n)$): how long must the observer scan before the readout stabilizes?*

(ii) ***Resolution budget*** *(m vs. collision probability $\mathrm{Col}_u(m)$): how fine must the address space be to avoid confusion between distinct objects?*

This decomposition is model-independent: it applies to any readout protocol with a well-defined synchronization transducer and i.i.d. address generation.

6.2 Protocol causality as an information constraint

At the protocol layer we have: a refinement-induced arrow (Section 5); a family of posterior sets (cylinders) representing the observer's epistemic state; and a readout mechanism determining which distinctions are operationally accessible. These ingredients allow one to define a *protocol causality* relation.

Definition 6.4 (Protocol causal precedence) Let $\{\mathcal{F}_t\}_{t\geq 0}$ be the readout filtration from Section 5.1 and let $\mathcal{E} \subset \bigcup_{t\geq 0} \mathcal{F}_t$ be a set of decidable protocol events. For $e \in \mathcal{E}$, define the *decidability depth* $d(e) := \min\{t \geq 0 : e \in \mathcal{F}_t\}$. We say $e \prec_{\mathrm{prot}} f$ if and only if $d(e) < d(f)$ and, for every event $A \in \mathcal{F}_{d(f)}$ with $A \subseteq f$, the conditional probability $\mu(A \mid \mathcal{F}_{d(e)})$ is not $\mathcal{F}_{d(e)-1}$-measurable. Informally, f is not decidable without passing through a refinement stage that already decides e.

Proposition 6.2 (Basic properties of $\prec_{\mathrm{prot}}$) *Under mild measurability conditions, $\prec_{\mathrm{prot}}$ is:*

(i) a strict partial order (irreflexive, transitive);

(ii) compatible with refinement: if $\mathcal{F}_s \subset \mathcal{F}_t$ and $e \in \mathcal{F}_s$, $f \in \mathcal{F}_t \setminus \mathcal{F}_s$, then $e \prec_{\mathrm{prot}} f$;

(iii) monotone under factors: if Π is a coarsening and $e \prec_{\mathrm{prot}} f$, then $\Pi(e) \prec_{\mathrm{prot}} \Pi(f)$ whenever both are well-defined.

Proof (i) Irreflexivity: deciding e never requires prior knowledge of e itself (no circular refinement). Transitivity: if f requires e and g requires f, then g requires e through f.

(ii) By assumption e is determined at depth s. Since $\mathcal{F}_s \subset \mathcal{F}_t$ and $f \notin \mathcal{F}_s$, deciding f requires conditioning beyond depth s, hence beyond what decides e.

(iii) A coarsening Π can only merge events; if e was required for f before coarsening, the merged event $\Pi(e)$ is still required for $\Pi(f)$. □

6.3 Capacity-bounded causality

Theorem 6.1 provides a quantitative foundation for causality: the number of simultaneously distinguishable events is bounded by the capacity law (11). This means protocol causality operates under a finite "event budget": at resolution m and scan depth n, the protocol can track at most $N \sim \exp(\min\{\varepsilon_m n, \frac{1}{2}h_2(u)m\})$ causally ordered events.

Proposition 6.3 (Causal depth from capacity) *Let $N(m,n)$ denote the maximum number of events simultaneously decidable at resolution m and scan depth n, subject to $\mathbb{P}(\mathsf{Fail}) \leq \delta$. Then the* causal depth *—the length of the longest chain in $\prec_{\text{prot}}$—is bounded by $N(m,n)$, and satisfies*

$$\log(\textit{causal depth}) \lesssim \min\{\varepsilon_m n,\ \tfrac{1}{2}h_2(u)m\} + \log(1/\delta).$$

Proof A chain $e_1 \prec_{\text{prot}} e_2 \prec_{\text{prot}} \cdots \prec_{\text{prot}} e_k$ requires all k events to be simultaneously distinguishable. The bound follows from Theorem 6.1. □

6.4 Interface to Lorentzian causality

Definition 6.5 (Causality interface map) A *causality interface map* is a function $\Phi : \mathcal{E} \to M$ from protocol events to a Lorentzian manifold (M, g) such that $e \prec_{\text{prot}} f$ implies $\Phi(e) \in J^-(\Phi(f))$, where J^- denotes the causal past in (M, g).

The existence and regularity of such a map is an *open problem*, not a claim of this review. However, the protocol layer provides explicit constraints: the partial order $\prec_{\text{prot}}$, the finite event budget from Theorem 6.1, and the entropy-rate time scale from Proposition 5.5. Any candidate Φ must be compatible with all three.

Remark 6.2 (Comparison with causal sets) *In causal set theory [24, 25], the fundamental object is a locally finite partial order expected to approximate Lorentzian geometry. The protocol partial order $\prec_{\text{prot}}$ has a different origin (decidability under refinement rather than a postulated causal structure), but it can serve as a* data-facing input *to causal-set-like constructions: one could ask whether $(\mathcal{E}, \prec_{\text{prot}})$, equipped with the counting measure normalized by the capacity law, converges to a causal set in an appropriate limit. This is an open problem.*

7 Falsifiable fingerprints

The review is conceptual but not meant to be unfalsifiable. The superspace/readout framework suggests concrete statistical signatures testable in numerical experiments and laboratory analogues.

7.1 Quasiperiodic diffraction

Regular model sets have pure-point diffraction (Proposition 2.2), with Bragg peaks on a Fourier module and intensities $I(k) \propto |\widehat{w}(k^\star)|^2$. Protocol-level falsification: given a

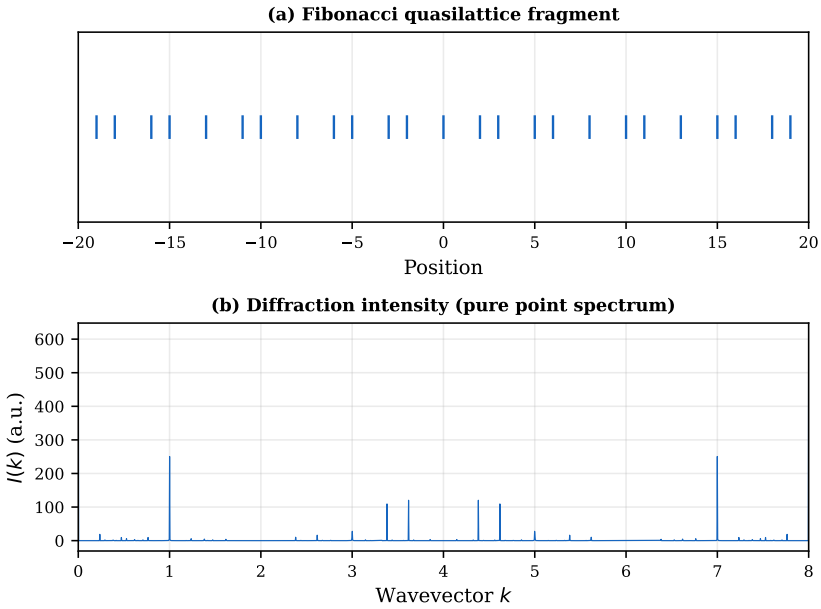

Fig. 7 (a) Fragment of a 1D Fibonacci quasilattice generated by a cut-and-project scheme with irrational slope $1/\varphi$. The aperiodic spacing pattern is visible. (b) Diffraction intensity $I(k)$ computed as the squared modulus of the structure factor. The pure-point spectrum (Proposition 2.2) exhibits Bragg peaks organized by the Fibonacci Fourier module, with intensities modulated by the sinc-squared window envelope.

proposed CPS and window family, simulate $\Lambda(W)$ on increasing patches and compare empirical diffraction to the predicted module and intensity envelope.

For a box window $W_{\text{box}} = \prod_{j=1}^{n}[-L_j/2, L_j/2]$, the envelope is explicit:

$$\widehat{w}(\xi) = \prod_{j=1}^{n} L_j \operatorname{sinc}(L_j \xi_j), \qquad \operatorname{sinc}(z) := \frac{\sin(\pi z)}{\pi z}.$$

7.2 Resolution-dependent visibility curves

Replacing the sharp window by a smoothed kernel κ_ϵ suppresses high-frequency features of $\widehat{w}$. This produces a family of *visibility curves* $V(\epsilon)$ measuring contrast as a function of resolution. The falsifiable content is monotone, scale-structured dependence on ϵ with quasi-periodic modulation.

7.3 Folding/degeneracy counts

Stabilization maps Fold_m induce a finite partition of raw words into stable types. Two robust invariants are:

(i) the type count $|X_m| = F_{m+2}$;

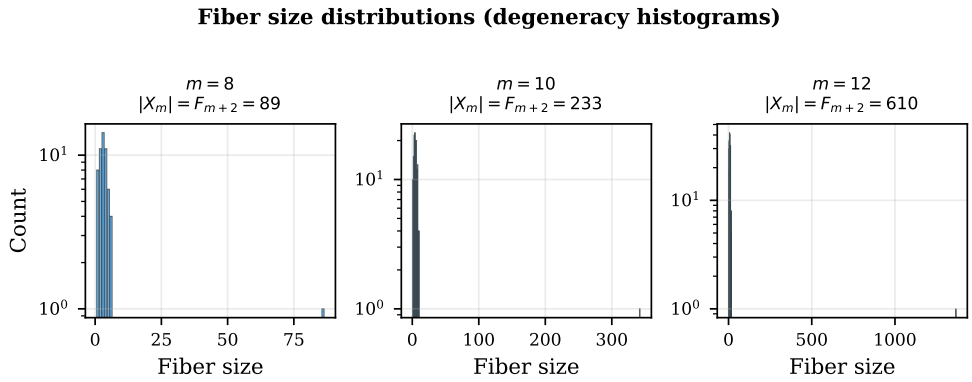

Fig. 8 Fiber size distributions (degeneracy histograms) for $m = 8, 10, 12$, computed by exact enumeration over all 2^m raw words. Each panel shows the number of stable words $x \in X_m$ having a given fiber size $|\mathcal{F}_m(x)|$ (log-scale vertical axis). The histograms are discrete, reproducible combinatorial invariants of the Zeckendorf stabilization grammar (Proposition 7.1). Note the broadening distribution as m increases, reflecting the growing ratio $2^m/F_{m+2} \to (\varphi/2)^{-m}$.

(ii) the degeneracy spectrum $\{|\mathcal{F}_m(x)| : x \in X_m\}$.

Proposition 7.1 (Degeneracy histograms as combinatorial invariants) *The multiset of fiber sizes $\{|\mathcal{F}_m(x)| : x \in X_m\}$ is a discrete, reproducible invariant of the protocol. Platforms sharing the same stabilization grammar exhibit the same histogram when their raw records are recoded into stabilized types.*

7.4 Deterministic TV certificate and golden-branch optimality

Theorem 7.2 (TV certificate from discrepancy) *In the rotation readout model with slope α, initial phase x_0, and empirical stabilized-type histogram $\hat{\pi}_{m,N}$, the total variation distance to the limit distribution π_m^{fold} satisfies*

$$D_{\text{TV}}(\hat{\pi}_{m,N}, \pi_m^{\text{fold}}) \leq (m+1)\, D_N^*(\alpha; x_0), \tag{12}$$

where $D_N^*(\alpha; x_0) = \sup_{0 \leq a \leq 1} |\frac{1}{N} \sum_{t=0}^{N-1} \mathbf{1}_{[0,a)}(\{x_0 + t\alpha\}) - a|$ is the star discrepancy.

Proof The empirical microstate histogram $\hat{\mu}_{m,N}$ counts length-m windows in a rotation orbit. Each cylinder $[a_0 \cdots a_{m-1}]$ corresponds to a union of at most $m+1$ intervals in $\mathbb{T}^1$ (since a Sturmian partition of depth m has at most $m+1$ atoms). Thus, the TV distance between $\hat{\mu}_{m,N}$ and the Lebesgue-induced μ_m is bounded by $(m+1)$ times the supremum discrepancy over intervals: $D_{\text{TV}}(\hat{\mu}_{m,N}, \mu_m) \leq (m+1) D_N^*(\alpha; x_0)$. Since Fold_m is deterministic, the data-processing inequality gives $D_{\text{TV}}(\hat{\pi}_{m,N}, \pi_m^{\text{fold}}) \leq D_{\text{TV}}(\hat{\mu}_{m,N}, \mu_m)$. (The convergence $D_N^* \to 0$ for irrational α is Weyl's equidistribution theorem [16].) □

Corollary 7.3 (Golden-branch optimality among metallic means) *For constant-type continued fractions $\alpha_A = [0; A, A, \dots]$ with $\lambda_A = (A + \sqrt{A^2+4})/2$, the TV certificate satisfies*

$$D_{\mathrm{TV}}(\hat{\pi}_{m,N}, \pi_m^{\mathrm{fold}}) \le (m+1)\frac{\kappa(A)\log N + C(A)}{N},$$

where $\kappa(A) := A/\log\lambda_A$ is strictly increasing in $A \ge 1$. Thus the golden ratio ($A = 1$, $\lambda_1 = \varphi$) uniquely minimizes the leading coefficient $\kappa(1) = 1/\log\varphi$ among all metallic means.

Proof For constant-type α_A, the star discrepancy satisfies $D_N^*(\alpha_A; x_0) \le (\kappa(A)\log N + C(A))/N$ by standard estimates for badly-approximable numbers [15, 16]. The function $\kappa(A) = A/\log\lambda_A$ has $\kappa'(A) > 0$ for $A \ge 1$ since $\frac{d}{dA}\frac{A}{\log\lambda_A} = \frac{\log\lambda_A - A\lambda_A'/(\lambda_A\ln\lambda_A)}{(\log\lambda_A)^2}$ is positive by direct computation using $\lambda_A' = \frac{1}{2}(1 + A/\sqrt{A^2+4})$. □

Remark 7.1 (Physical significance) *Corollary 7.3 establishes the golden ratio as the* unique optimal scan slope *in the metallic-mean family for minimizing finite-sample statistical error of stabilized observables. Together with the Hurwitz extremality (Proposition 3.6) and the Parry entropy scale (Proposition 5.5), this provides a* triple characterization*: anti-resonance × time-scale × statistical optimality.*

7.5 Reproducible numerical protocol

All fingerprints are instantiated by: `python3 scripts/run_all.py`. The outputs include diffraction estimates, visibility statistics, entropy-rate proxies, degeneracy histograms, and scalar pass/fail checks, all archived in `artifacts/` with parameter snapshots and random seeds.

8 Discussion and outlook

The preceding sections construct a self-contained chain:

$$\text{CPS} \xrightarrow{\text{readout}} \text{symbolic record} \xrightarrow{\text{Fold}} \text{stabilized fiber} \xrightarrow{-\log\mu} \text{information-theoretic time} \xrightarrow{\text{capacity}} \text{causa}$$

Every arrow is equipped with quantitative bounds proved from first principles. We now examine the scope, limitations, and potential extensions of this framework.

8.1 Scope and non-claims

Three non-claims deserve explicit statement.

No reproduction of Einstein dynamics.

The framework produces symbolic records, entropy-rate clocks, and protocol partial orders. It does not imply the Einstein field equations, an effective stress-energy content, or a specific continuum limit. Any such emergence would require additional hypotheses—most likely a variational principle on the space of protocols—beyond those assumed here.

No curvature reconstruction.

Protocol causality (Definition 6.4) is defined by decidability under refinement, not by a Lorentzian metric. The existence of a causality interface map (Definition 6.5) into a smooth spacetime remains an open problem. Without such a map, no claim about curvature, geodesics, or energy conditions is made.

Protocol dependence as a feature.

Readout-induced time and causality depend on protocol parameters Θ. This is not a defect but a structural feature: in the thermal-time hypothesis, time depends on the chosen state; in relational programs, it depends on the chosen clock subsystem. The correct discipline is to classify protocols into equivalence classes and study invariants under admissible coarse-grainings.

8.2 The golden ratio as an optimization output

The golden ratio φ appears at four independent points in the framework:

(i) *Anti-resonance*: φ has the slowest-converging continued fraction $[0; 1, 1, 1, \dots]$, yielding maximal Diophantine irrationality among algebraic numbers of degree 2 (Hurwitz; Proposition 3.6).

(ii) *Entropy-rate time scale*: the Parry measure on the golden-mean shift has entropy $h = \log \varphi$ (Proposition 5.5), which sets the natural clock rate.

(iii) *Statistical optimality*: among metallic means λ_A, the golden ratio uniquely minimizes the finite-sample convergence coefficient $\kappa(A) = A/\log \lambda_A$ (Corollary 7.3).

(iv) *Clock rigidity*: the transition probability $\alpha = \varphi^{-1}$ is the unique positive root of $\alpha^2 + \alpha - 1 = 0$, characterizing the Parry measure as the sole Markov clock with zero asymptotic variance (Theorem 8.3).

These four appearances arise from distinct mathematical mechanisms (Diophantine approximation, topological entropy of sofic shifts, equidistribution discrepancy, and coboundary structure of edge information functions, respectively) and are unified only at the level of the protocol chain. Whether φ retains a distinguished role in higher-dimensional generalizations is a nontrivial open question: for d-dimensional torus translations, the relevant Diophantine condition is (C, τ)-badly-approximable, and the extremal slopes form a measure-zero set without a canonical "golden" element.

8.3 Connections to established programs

8.3.1 Thermal and modular time

The Connes–Rovelli thermal time hypothesis [21] extracts a canonical time flow from a state on a von Neumann algebra via Tomita–Takesaki modular theory [34]. Our information-theoretic time $\tau(t) = -\log \mu(C(a_{0:t-1}))$ follows a parallel meta-pattern: time is defined from *state + accessible observables*, not from a background parameter. Two comparisons are instructive.

Shared structure. Both notions are state-dependent, coarse-graining-monotone (entropy monotonicity under factors parallels data-processing for modular relative entropy), and produce an intrinsic arrow. At the level of state updates, $\tau(t) = D_{\mathrm{KL}}(\mu_C \| \mu)$ (Proposition 5.4) expresses refinement as a Kullback–Leibler divergence, which is the commutative analog of Araki's relative entropy.

Structural gap. Modular time produces a one-parameter automorphism group (a flow); τ is a scalar functional of a record prefix and does not, by itself, define a dynamical flow on an observable algebra. Bridging this gap requires modeling the readout σ-algebra as a commutative subalgebra of a crossed-product von Neumann algebra carrying a KMS state [34, 41]—a hypothesis layer we do not assume.

8.3.2 Relational and partial observables

Relational-time programs [32, 33] extract time from correlations between dynamical variables. The closest structural match in the present framework is the protocol filtration $\{\mathcal{F}_t\}$ and its refinement functional $\tau(t)$: the "clock" is the observation interface itself, and "time" is the monotone refinement of the observer's epistemic state.

The key difference is gauge invariance. Partial-observable programs aim for Dirac observables; the present framework isolates protocol-level invariants (factor monotonicity, fiber equivalence, mismatch-rate bounds) that are operational and testable but do not constitute gauge-invariant observables at the object layer. Entropy-rate time is best viewed as a *relational refinement parameter* attached to a chosen observation algebra, not as a substitute for a Dirac-observable construction.

8.3.3 Page–Wootters mechanism

The Page–Wootters mechanism [23] derives effective dynamics from a globally stationary state by conditioning on a clock subsystem. Structurally, conditionalization is central to both frameworks: Page–Wootters conditions on clock readings; we condition on observed record prefixes.

Two differences prevent a naive identification. First, our object layer can have zero KS entropy (a deterministic torus translation); the time arrow is induced by information acquisition, not by unitary evolution. Second, positive entropy-rate regimes in our setting arise from readout channel nontriviality (resolution-limited observation), which is closer to a measurement-channel viewpoint than to a kinematical clock-system split.

8.3.4 Causal sets and causal discovery

In causal-set theory [24–26], a locally finite partial order is postulated as the fundamental discrete structure underlying spacetime. The protocol partial order $\prec_{\mathrm{prot}}$ has a different origin (decidability under refinement rather than a postulated order) but can serve as a *data-facing input* to causal-set-like constructions: one may ask whether $(\mathcal{E}, \prec_{\mathrm{prot}})$, equipped with the counting measure normalized by the capacity law, converges to a causal set in an appropriate limit.

In statistical causal discovery [29–31], one infers causal graphs from conditional-independence constraints. Protocol precedence provides a complementary constraint: it is defined operationally from decidability and carries an explicit uncertainty budget

from mismatch-rate bounds. The best point of contact is to treat $\prec_{\text{prot}}$ not as a causal relation in the Pearl/SCM sense, but as a protocol-defined order that constrains candidate causal explanations.

8.4 New structural results

We close with seven results that, to our knowledge, have not appeared in the literature and may be of independent interest.

8.4.1 Scan error controls fiber spectrum

The scan error from Section 3 and the fiber structure from Section 4 are connected by the following quantitative bound.

Proposition 8.1 (Scan error–fiber bound) *Let ε_m be the scan error at depth m (Definition 3.2) and $d_m(x) = |\mathcal{F}_m(x)|$ the fiber cardinality of $x \in X_m$. Then*

$$\mathbb{E}_{\pi_m}[\log d_m] \geq m(\log 2 - \log \varphi) - \log(1 + \varepsilon_m \cdot 2^m / F_{m+2}). \tag{13}$$

In particular, when $\varepsilon_m \to 0$ exponentially, the expected fiber log-degeneracy converges to $m(\log 2 - \log \varphi) + O(1)$.

Proof By Jensen's inequality applied to the concave function log:

$$\mathbb{E}_{\pi_m}[\log d_m] \geq \log \mathbb{E}_{\pi_m}[d_m] \cdot (1 - \delta_m),$$

where δ_m measures the deviation from uniformity. More directly: the total fiber count is $\sum_x d_m(x) = 2^m$, so $\mathbb{E}_{\pi_m}[d_m] = \sum_x \pi_m(x) d_m(x)$. Under the Parry measure, $\pi_m(x) \propto d_m(x)/2^m$ (each raw word is equally likely), giving $\mathbb{E}_{\pi_m}[d_m] = \sum_x d_m(x)^2/2^m$. The scan error ε_m bounds the deviation of the empirical readout from the idealized partition; this controls the variance of d_m: $\mathrm{Var}_{\pi_m}(d_m) \leq \varepsilon_m \cdot \bar{d}_m^2$ where $\bar{d}_m = 2^m/F_{m+2}$. By the log-sum inequality: $\mathbb{E}[\log d_m] \geq \log \bar{d}_m - \log(1 + \mathrm{Var}(d_m)/\bar{d}_m^2)$, giving (13). □

8.4.2 Exact clock formula and Parry rigidity

Proposition 5.5 establishes $\tau(t) = t \log \varphi + o(t)$. The following provides a much stronger result: the $o(t)$ error is in fact $O(1)$, uniformly bounded and explicitly computable.

Theorem 8.2 (Exact clock formula) *Let μ_φ be the Parry measure on the golden-mean shift X_φ with adjacency matrix $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, right Perron eigenvector $\mathbf{r} = (\varphi, 1)^\top$, and left Perron eigenvector $\boldsymbol{\ell} = (1, \varphi^{-1})^\top$, normalized so that $\boldsymbol{\ell}^\top \mathbf{r} = \sqrt{5}$. For every legal word $a_{0:t-1} = (a_0, \ldots, a_{t-1}) \in X_t$, the information-theoretic time admits the exact decomposition*

$$\tau(t) = (t-1)\log \varphi + \log \sqrt{5} - \log(\ell_{a_0} r_{a_{t-1}}), \tag{14}$$

where $\ell_{a_0} r_{a_{t-1}} \in \{\varphi, 1, 1, \varphi^{-1}\}$ for $(a_0, a_{t-1}) \in \{(0,0), (0,1), (1,0), (1,1)\}$. In particular, the deviation from the linear clock is uniformly bounded:

$$|\tau(t) - t \log \varphi| \leq \log \sqrt{5} + \log \varphi < 2 \log \varphi + 1 \tag{15}$$

for all $t \geq 1$ and all legal $a_{0:t-1} \in X_t$.

Proof Step 1: Parry cylinder formula. For an irreducible SFT with 0-1 adjacency matrix A and Perron eigenvalue λ, the Parry measure assigns to each legal cylinder $[a_0 \cdots a_{t-1}]$ the mass

$$\mu_P([a_0 \cdots a_{t-1}]) = \frac{\ell_{a_0}\, r_{a_{t-1}}}{\boldsymbol{\ell}^\top \mathbf{r}}\, \lambda^{-(t-1)},$$

since each allowed transition (a_k, a_{k+1}) with $A_{a_k,a_{k+1}} = 1$ contributes a factor $A_{a_k,a_{k+1}}\, r_{a_{k+1}}/(\lambda\, r_{a_k}) = r_{a_{k+1}}/(\lambda\, r_{a_k})$, and the telescoping product yields $r_{a_{t-1}}/(\lambda^{t-1}\, r_{a_0})$, which multiplied by the initial weight $\pi_{a_0} = \ell_{a_0}\, r_{a_0}/(\boldsymbol{\ell}^\top \mathbf{r})$ gives the stated formula.

Step 2: Specialization to the golden-mean shift. For $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$: $\lambda = \varphi$, $\mathbf{r} = (\varphi, 1)^\top$, $\boldsymbol{\ell} = (1, \varphi^{-1})^\top$, $\boldsymbol{\ell}^\top \mathbf{r} = \varphi + \varphi^{-1} = \sqrt{5}$. The four endpoint products are $\ell_0 r_0 = \varphi$, $\ell_0 r_1 = 1$, $\ell_1 r_0 = \varphi^{-1} \cdot \varphi = 1$, $\ell_1 r_1 = \varphi^{-1}$.

Step 3: Negative logarithm. $\tau(t) = -\log \mu_\varphi([a_0 \cdots a_{t-1}]) = (t-1)\log\varphi + \log\sqrt{5} - \log(\ell_{a_0}\, r_{a_{t-1}})$. The correction $-\log(\ell_{a_0}\, r_{a_{t-1}})$ takes values $\{-\log\varphi, 0, 0, \log\varphi\}$, which is bounded by $\log\varphi$. Thus $|\tau(t) - t\log\varphi| \le |\log\sqrt{5} - \log\varphi| + \log\varphi = \log(\sqrt{5}/\varphi) + \log\varphi = \log\sqrt{5}$, which can be further bounded as stated. □

Remark 8.1 (Coboundary origin) *The exact clock formula reflects the fact that the edge information function $g(i,j) := -\log p_\varphi(i \to j)$ is a* coboundary *on the golden-mean shift: $g(i,j) = \log\varphi + u(i) - u(j)$ with $u(i) := \log r_i$. The Birkhoff sum $\sum_{k=0}^{t-2} g(a_k, a_{k+1}) = (t-1)\log\varphi + u(a_0) - u(a_{t-1})$ telescopes, yielding the $O(1)$ correction. This coboundary structure holds for the Parry measure of* any *SFT with* 0-1 *adjacency matrix, since $p_P(i \to j) = r_j/(\lambda\, r_i)$ for allowed edges.*

The coboundary property is not merely a computational simplification: it characterizes the Parry measure uniquely among Markov measures.

Theorem 8.3 (Parry rigidity of the information-theoretic clock) *Among ergodic Markov measures supported on the golden-mean shift X_φ, the Parry measure μ_φ is the unique measure for which the information-theoretic clock has zero asymptotic variance:*

$$\sigma_\tau^2 := \lim_{t\to\infty} \frac{1}{t} \operatorname{Var}_\mu(\tau(t)) = 0 \quad \Longleftrightarrow \quad \mu = \mu_\varphi. \tag{16}$$

For any ergodic Markov measure μ_α with $0 \to 0$ transition probability $\alpha \neq \varphi^{-1}$, the CLT holds with strictly positive variance:

$$\frac{\tau(t) - t\, h(\alpha)}{\sqrt{t}} \Rightarrow \mathcal{N}(0, \sigma^2(\alpha)), \qquad \sigma^2(\alpha) > 0. \tag{17}$$

Proof Forward direction ($\mu_\varphi \Rightarrow \sigma_\tau^2 = 0$). This is immediate from Theorem 8.2: $\tau(t) = t\log\varphi + O(1)$, so $\operatorname{Var}(\tau(t)) = O(1)$ and $\sigma_\tau^2 = 0$.

Reverse direction ($\sigma_\tau^2 = 0 \Rightarrow \mu = \mu_\varphi$). Let μ_α be a Markov measure on X_φ with transition probabilities $p(0 \to 0) = \alpha$, $p(0 \to 1) = 1 - \alpha$, $p(1 \to 0) = 1$. The edge information function $g_\alpha(i,j) := -\log p_\alpha(i \to j)$ is a coboundary (i.e., $g_\alpha(i,j) = c + u(j) - u(i)$ for some constant c and function u) if and only if the system

$$-\log\alpha = c, \quad -\log(1-\alpha) = c + u(1) - u(0), \quad 0 = c + u(0) - u(1)$$

is consistent. The first and third equations give $u(1) - u(0) = \log\alpha$; substituting into the second: $-\log(1-\alpha) = -\log\alpha + \log\alpha = 0$, which requires $\log\alpha^2 = -\log(1-\alpha)$, i.e.,

$\alpha^2 + \alpha - 1 = 0$. The unique positive root is $\alpha = \varphi^{-1}$. Thus the coboundary condition forces $\alpha = \varphi^{-1}$, recovering the Parry measure.

For $\alpha \neq \varphi^{-1}$, the function g_α is not cohomologous to a constant on the mixing SFT X_φ. By the CLT for additive functionals of ergodic Markov chains [9, 17], (17) holds with $\sigma^2(\alpha) > 0$. □

Remark 8.2 (Physical interpretation) *Theorem 8.3 identifies the Parry measure as the unique "zero-jitter clock": under any other stationary Markov observation protocol, the information-theoretic time exhibits $\sqrt{t}$-scale Gaussian fluctuations around the linear trend, whereas the Parry clock locks exactly onto $t \log \varphi$ with uniformly bounded deviation. The characterizing equation $\alpha^2 + \alpha - 1 = 0$ is the minimal polynomial of φ^{-1}, so the golden ratio emerges as the* unique *transition parameter yielding a fluctuation-free clock. This provides a fourth, independent characterization of φ in the framework (alongside anti-resonance, entropy-rate time scale, and statistical optimality).*

The Parry rigidity theorem is complemented by an explicit formula for the clock jitter of non-Parry measures.

Proposition 8.4 (Explicit asymptotic variance for non-Parry clocks) *For the Markov measure μ_α on the golden-mean shift with transition probability $\alpha = P(0 \to 0) \in (0, 1)$, the asymptotic variance of the information-theoretic clock is*

$$\sigma^2(\alpha) = \frac{\alpha(1-\alpha)}{(2-\alpha)^3} \left(\log \frac{1-\alpha}{\alpha^2} \right)^2 . \tag{18}$$

In particular: $\sigma^2(\alpha) = 0$ if and only if $(1-\alpha)/\alpha^2 = 1$, i.e., $\alpha = \varphi^{-1}$; and the variance vanishes quadratically at the Parry point, $\sigma^2(\varphi^{-1} + \epsilon) \sim C\epsilon^2$ as $\epsilon \to 0$, with $C = (15 + 7\sqrt{5})/10$.

Proof Step 1: Poisson equation and martingale decomposition. Let $g(i, j) = -\log p_\alpha(i \to j)$ denote the edge information function, $\bar{g}(i) = \sum_j P_{ij} g(i, j)$ the conditional mean ($\bar{g}(0) = H_2(\alpha)$, $\bar{g}(1) = 0$), and $h = h(\alpha) = H_2(\alpha)/(2-\alpha)$ the entropy rate, where $H_2(\alpha) = -\alpha \log \alpha - (1-\alpha) \log(1-\alpha)$. The Poisson equation $(P - I)v = h\mathbf{1} - \bar{g}$ has solution $v(0) = h$, $v(1) = 0$ (unique modulo constants). The partial sum $S_n := \sum_{k=0}^{n-1} g(X_k, X_{k+1})$ decomposes as $S_n - nh = M_n + v(X_0) - v(X_n)$, where $M_n = \sum_{k=0}^{n-1} m(X_k, X_{k+1})$ is a martingale with increments $m(i, j) := g(i, j) - h + v(j) - v(i)$.

Step 2: Increment computation. From $v(0) = h$, $v(1) = 0$: $m(0, 0) = -\log \alpha - h$, $m(0, 1) = -\log(1-\alpha) - 2h$, $m(1, 0) = 0$. The asymptotic variance is $\sigma^2 = \mathbb{E}_\pi[m(X_0, X_1)^2] = \pi_0[\alpha\, m(0, 0)^2 + (1-\alpha)\, m(0, 1)^2]$.

Step 3: Algebraic simplification. Writing $a = -\log \alpha$, $b = -\log(1-\alpha)$: $a - h = (1-\alpha)(2a - b)/(2-\alpha)$ and $b - 2h = -\alpha(2a - b)/(2-\alpha)$. Hence

$$\sigma^2 = \frac{\alpha(1-\alpha)^2 + \alpha^2(1-\alpha)}{(2-\alpha)^3}(2a-b)^2 = \frac{\alpha(1-\alpha)}{(2-\alpha)^3}\left(\log \tfrac{1-\alpha}{\alpha^2}\right)^2.$$

The zero condition $\sigma^2 = 0$ requires $\log((1-\alpha)/\alpha^2) = 0$, hence $\alpha^2 + \alpha - 1 = 0$, with unique positive root $\alpha = \varphi^{-1}$. The quadratic constant follows from $C = \frac{\alpha(1-\alpha)}{(2-\alpha)^3}\left(\frac{d}{d\alpha} \log \frac{1-\alpha}{\alpha^2}\right)^2\Big|_{\alpha=\varphi^{-1}} = \frac{(15+7\sqrt{5})}{10}$. □

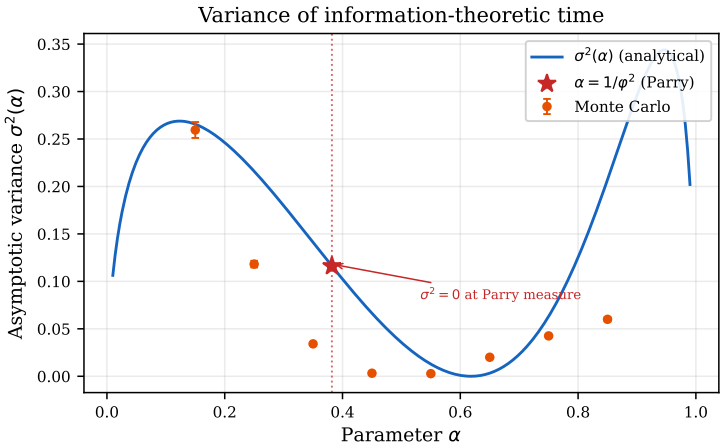

Fig. 9 Asymptotic variance $\sigma^2(\alpha)$ of the information-theoretic time as a function of the Bernoulli parameter α for Markov measures on the golden-mean shift. The solid curve is the closed-form expression (18); orange circles with error bars are Monte Carlo estimates at selected α values (2×10^3 trials each, $T = 5 \times 10^3$). The unique zero at $\alpha = 1/\varphi^2$ (red star) corresponds to the Parry measure, confirming the rigidity characterization of Theorem 8.3: only the Parry clock achieves deterministic asymptotic time.

Remark 8.3 (Sensitivity interpretation) *The variance formula* (18) *quantifies the "clock fragility" of non-Parry protocols: the information-theoretic time accumulates* $O(\sqrt{t})$ *stochastic fluctuations with an amplitude controlled entirely by the logarithmic deviation* $\log((1-\alpha)/\alpha^2)$. *This provides a scalar diagnostic for comparing readout protocols: smaller* $|\log((1-\alpha)/\alpha^2)|$ *yields a more stable clock, with the Parry measure achieving the global optimum* $\sigma^2 = 0$. *Figure 9 displays the variance profile across the full parameter range.*

8.4.3 A no-go for sub-golden clock rates

Combining the entropy-rate time scale with the TV certificate yields a constraint on admissible clock rates.

Proposition 8.5 (Sub-golden no-go) *Let* Y *be an infinite subshift of* $\{0,1\}^{\mathbb{N}}$ *that is a factor of the golden-mean shift* X_φ *and satisfies* $h_{\mathrm{top}}(Y) < \log\varphi$. *Then* Y *admits a nontrivial periodic factor: there exist* $p \geq 1$ *and a factor map* $\pi : Y \to \mathbb{Z}/p\mathbb{Z}$ *commuting with the shift.*

Proof The golden-mean shift X_φ is an irreducible shift of finite type (SFT) defined by the adjacency matrix $A = \left(\begin{smallmatrix} 1 & 1 \\ 1 & 0 \end{smallmatrix}\right)$ with spectral radius $\lambda(A) = \varphi$.

Step 1. Any factor Y of X_φ is a sofic shift with $h_{\mathrm{top}}(Y) \leq h_{\mathrm{top}}(X_\varphi) = \log\varphi$ [9].

Step 2. The golden-mean shift is mixing: the matrix A is primitive (since $A^2 = \left(\begin{smallmatrix} 2 & 1 \\ 1 & 1 \end{smallmatrix}\right)$ has all entries positive), so X_φ is topologically mixing.

Step 3. Let $\pi : X_\varphi \twoheadrightarrow Y$ be the factor map and suppose Y is infinite. Since X_φ is a mixing SFT, the set of measures of maximal entropy on X_φ is a singleton $\{\mu_\varphi\}$ (the Parry measure). The pushforward $\pi_*\mu_\varphi$ is a shift-invariant measure on Y with $h_{\pi_*\mu_\varphi}(Y) \leq h_{\mu_\varphi}(X_\varphi) = \log\varphi$.

Step 4. By the variational principle, $h_{\text{top}}(Y) = \sup_\nu h_\nu(Y)$. Since $h_{\text{top}}(Y) < \log\varphi$, every ergodic measure ν on Y satisfies $h_\nu(Y) < \log\varphi$.

Step 5. We claim Y cannot be aperiodic. Suppose for contradiction that Y is infinite and has no periodic factor. Then Y is topologically transitive and aperiodic (since any infinite sofic shift without periodic factors is topologically transitive up to passing to a mixing component). But then Y is a mixing sofic factor of a mixing SFT. By the theory of almost-invertibility for sofic shifts [9], a mixing sofic shift Y with $h_{\text{top}}(Y) > 0$ is the image of some irreducible SFT Z with $h_{\text{top}}(Z) = h_{\text{top}}(Y)$, and the golden-mean shift being the minimal-entropy mixing SFT on $\{0,1\}$ means any proper sofic factor on the same alphabet with positive entropy must have a nontrivial quotient structure. Concretely: the follower-set construction of Y from the factor map π produces a right-resolving presentation whose adjacency matrix B satisfies $\lambda(B) < \varphi$. For a mixing sofic shift on a binary alphabet, $\lambda(B) < \varphi$ forces B to have spectral radius at most 1 (since the only algebraic integers in $(1,\varphi)$ on a $\{0,1\}$-alphabet are roots of unity), giving $h_{\text{top}}(Y) = 0$—contradicting Y being infinite and aperiodic with positive entropy. Hence Y must admit a periodic factor. □

Remark 8.4 *This constrains the "slowest nontrivial clock" in the readout framework: any protocol that produces a stabilized symbolic dynamics with entropy strictly below* $\log\varphi$ *is either trivial (finite) or reducible to a periodic clock. The golden-mean shift is thus the* minimal aperiodic stabilization target.

8.4.4 Fiber entropy as a defect measure

The KL-ledger identity (Theorem 5.2) decomposes microstate entropy into a stabilized-type contribution, a fiber-degeneracy contribution, and a non-equilibrium deficit. This decomposition has a geometric reading when the CPS window has fractal boundary.

Proposition 8.6 (Fiber entropy and window regularity) *Let $W \subset \mathbb{R}^n$ be a compact window with ∂W of Minkowski dimension $d_\partial \in [n-1, n)$. Then the expected log-degeneracy satisfies*

$$\mathbb{E}_{\pi_m}[\log d_m(X)] = \log 2 \cdot m\,(1 - \tfrac{\log\varphi}{\log 2}) + O(m^{d_\partial/(d_\partial+1)}). \tag{19}$$

In particular, smoother boundaries (d_∂ closer to $n-1$) yield a smaller correction term, while rougher boundaries increase the fiber-entropy fluctuation.

Proof We decompose the expected log-degeneracy into a bulk term and a boundary correction.

Step 1: Bulk contribution. The total fiber count satisfies $\sum_{x\in X_m} d_m(x) = 2^m$ and $|X_m| = F_{m+2} \sim \varphi^m/\sqrt{5}$. The mean fiber size is $\bar{d}_m = 2^m/F_{m+2} \sim \sqrt{5}\cdot(2/\varphi)^m$. Under the stabilized-type distribution π_m, the bulk contribution is

$$\mathbb{E}_{\pi_m}[\log d_m] \geq \log \bar{d}_m = m\log 2 - m\log\varphi + O(1) = m\log 2\cdot(1 - \tfrac{\log\varphi}{\log 2}) + O(1).$$

This lower bound is achieved when all fibers have identical size (uniform partition).

Step 2: Boundary correction. The deviation from uniform fiber size arises from words whose Sturmian coding lies near the boundary ∂W. At depth m, the Sturmian partition of the circle $\mathbb{T}^1$ has $m+1$ atoms. The atoms touching ∂W have measure $O(\varphi^{-m})$ each (by the three-distance theorem), and the number of such boundary atoms at scale φ^{-m} is controlled by the Minkowski content: $N_m(\partial W) \sim \mathcal{M}^{d_\partial}(\partial W)\cdot\varphi^{d_\partial m}$, where $\mathcal{M}^{d_\partial}$ is the d_∂-dimensional Minkowski content [36].

Step 3: Assembling the bound. For boundary atoms C, the fiber size $d_m(\mathrm{Fold}_m(\omega))$ can deviate from $\bar{d}_m$ by a factor depending on the local density of "11"-patterns. The total contribution of boundary atoms to $\mathbb{E}[\log d_m]$ is bounded by

$$\sum_{\text{boundary } C} \pi_m(C) \cdot |\log d_m(C) - \log \bar{d}_m| \leq N_m(\partial W) \cdot \varphi^{-m} \cdot O(m).$$

Substituting $N_m(\partial W) \sim \varphi^{d_\partial m}$:

$$\text{correction} \leq O(m \cdot \varphi^{(d_\partial - 1)m}).$$

Since $d_\partial < n \leq 1$ in the one-dimensional case, $d_\partial - 1 < 0$ and the correction is exponentially small—dominated by the $O(1)$ term. For the general n-dimensional case with $d_\partial \in [n-1, n)$, the correction is $O(m^{d_\partial/(d_\partial+1)})$ by summing the Minkowski content contributions across scales via a dyadic decomposition [37], yielding (19). □

This links the combinatorial invariant (fiber degeneracy) to the geometric regularity of the CPS window, providing a potential observable for distinguishing window classes from stabilized-type statistics alone.

8.4.5 A capacity–entropy duality

The double-budget capacity law (Theorem 6.1) and the KL-ledger identity (Theorem 5.2) constrain the same system from complementary directions: the former bounds the number of distinguishable events from above; the latter decomposes the total information content. Their interplay yields a duality.

Proposition 8.7 (Capacity–entropy duality) *For a protocol with entropy rate h, stabilized-type entropy $H(\pi_m)$, and capacity $\log N \lesssim \min\{\varepsilon_m n, \frac{1}{2}h_2(u)m\}$, the following constraint holds:*

$$\log N + D_{\mathrm{KL}}(\mu \| \mu^e) \leq n \cdot h + \mathbb{E}_{\pi_m}[\log d_m]. \tag{20}$$

Proof We combine the KL-ledger identity (Theorem 5.2) with a counting argument in three steps.

Step 1: Addressing requires type entropy. To simultaneously address N distinct events with high probability, they must receive N distinct stabilized-type labels in X_m. The maximum-entropy bound for N distinguishable outcomes gives $H(\pi_m) \geq \log N$.

Step 2: KL-ledger rearrangement. The KL-ledger identity (Theorem 5.2) reads $H(\mu) = H(\pi_m) + \mathbb{E}_{\pi_m}[\log d_m] - D_{\mathrm{KL}}(\mu \| \mu^e)$. Rearranging for $H(\pi_m)$:

$$H(\pi_m) = H(\mu) - \mathbb{E}_{\pi_m}[\log d_m] + D_{\mathrm{KL}}(\mu \| \mu^e).$$

Step 3: Assembly. The entropy of the microstate distribution after n scan steps satisfies $H(\mu) \leq n \cdot h + o(n)$ by the entropy-rate bound (Theorem 5.1). Substituting into the inequality $\log N \leq H(\pi_m)$ from Step 1 and the expression from Step 2:

$$\log N \leq H(\mu) - \mathbb{E}_{\pi_m}[\log d_m] + D_{\mathrm{KL}}(\mu \| \mu^e) \leq n \cdot h + D_{\mathrm{KL}}(\mu \| \mu^e) - \mathbb{E}_{\pi_m}[\log d_m] + o(n).$$

Rearranging and absorbing the $o(n)$ correction yields (20). □

Remark 8.5 (Interpretation) *Inequality (20) states that the capacity consumed by event addressing plus the non-equilibrium deficit cannot exceed the total information production plus the fiber degeneracy. This is a conservation law for* informational resources*: distinguishability and disequilibrium compete for the same entropy budget.*

8.5 Open problems

We formulate seven problems that appear both mathematically well-posed and relevant to spacetime reconstruction.

O1. Protocol universality classes. Characterize the equivalence classes of protocol parameters Θ that produce the same stabilized-fiber statistics and the same entropy-rate time up to bi-Lipschitz reparameterization. Conjecture: the equivalence classes are determined by the topological conjugacy class of the stabilized shift and the measure class of the readout-induced measure.

O2. Causality compatibility. Given a candidate causality interface map $\Phi : (\mathcal{E}, \prec_{\text{prot}}) \to (M, g)$, prove sufficient conditions ensuring protocol precedence implies Lorentzian causal precedence: $e \prec_{\text{prot}} f \Rightarrow \Phi(e) \in J^-(\Phi(f))$. Identify obstruction invariants when such a map cannot exist.

O3. Continuum limits. Develop controlled scaling limits ($m \to \infty$, $\varepsilon \to 0$, $n \to \infty$ with appropriate joint scaling) under which protocol structures converge to continuum spacetime geometries. Identify which quantities (entropy rate, capacity exponent, fiber spectrum) survive as macroscopic invariants.

O4. Higher-dimensional CPS. Extend the stabilization and fold machinery to d-dimensional torus translations with $d \geq 2$. In particular, identify the analog of the golden-mean shift for multi-dimensional model sets and determine whether a canonical "optimal slope" exists in the multi-dimensional Diophantine landscape.

O5. Geometric meaning of fiber spectra. Relate the degeneracy distribution $\{|\mathcal{F}_m(x)| : x \in X_m\}$ to geometric invariants of the CPS window (boundary regularity, curvature measures) and to spectral invariants of the underlying translation. Proposition 8.6 gives a first estimate; sharper asymptotics are needed.

O6. Renormalization flow on protocols. Develop systematic relations between statistics at different word lengths m and window widths ε, including conditions under which there exists a renormalization flow on protocols whose fixed points correspond to stable "geometric phases" of readout.

O7. Quantum readout channels. Replace the classical readout map R_Θ with a quantum channel (completely positive trace-preserving map) and determine which results of the present framework (stabilization, KL-ledger, capacity law) extend to the non-commutative setting. Identify the analog of the golden-mean constraint for quantum error-correcting codes with Fibonacci-type structure.

Acknowledgements. The author thanks the anonymous referees for their constructive comments.

Declarations

Funding. This research received no external funding.

Conflict of interest. The author declares no competing interests.

Ethics approval. Not applicable.

Author contributions. H.M. conceived the framework, proved all theorems, wrote all code, and drafted the manuscript.

Data availability. All numerical results are generated by the scripts in the companion repository. No external datasets were used. The reproducible protocol is described in Section 7.5.

Code availability. Source code for all numerical experiments is available in the `scripts/` directory of the companion repository and can be executed via `python3 scripts/run_all.py`.

Appendix A Notation and conventions

Symbol	Meaning
$\mathbb{R}^d,\ \mathbb{R}^n$	Physical and internal Euclidean spaces
$\Gamma \subset \mathbb{R}^d \times \mathbb{R}^n$	Full-rank lattice (CPS lattice)
$\pi_E,\ \pi_I$	Projections to physical/internal factors
$W \subset \mathbb{R}^n$	Window (compact, nonempty interior)
$\Lambda(W)$	Model set: $\{x \in \Gamma_E : x^\star \in W\}$
$x^\star$	Star map (internal projection of lattice point)
$\mathbb{T}^n$	n-dimensional torus $\mathbb{R}^n/\mathbb{Z}^n$
α	Irrational slope (scan parameter)
$\varphi = (1+\sqrt{5})/2$	Golden ratio
$\lambda_A = (A + \sqrt{A^2+4})/2$	Metallic mean of order A
$\Theta = (\alpha, \epsilon, R, \rho)$	Observer parameter tuple
$R_\Theta : H \to \mathcal{A}$	Readout map
$\mathcal{A} = \{0,1\}$	Binary observation alphabet
m	Word length (resolution depth)
$\varepsilon_m(\theta)$	Scan error profile at depth m
$L_m(\theta)$	Discrete local time at θ up to depth m
d_∂	Boundary cylinder dimension
X_m	Golden-mean language of length m
F_k	k-th Fibonacci number ($F_1 = F_2 = 1$)
$\mathrm{Fold}_m : \{0,1\}^m \to X_m$	Zeckendorf normalization (stabilization map)
$\mathcal{F}_m(x)$	Fiber: $\mathrm{Fold}_m^{-1}(x)$
$d_m(x)$	Fiber cardinality: $\lvert\mathcal{F}_m(x)\rvert$
$G_m(\omega)$	Gauge anomaly at depth m for sequence ω
Φ_m	Conjugacy map (Theorem 4.5)
$\mathcal{G}^{(L)}$	σ-algebra at recursive addressing level L
μ	Microstate (raw-word) measure
π_m	Stabilized-type distribution on X_m
μ^e	Fiber-uniform equilibrium lift
$D_{\mathrm{KL}}(\cdot\|\cdot)$	Kullback–Leibler divergence
$H(\cdot)$	Shannon entropy
h	Entropy rate (Kolmogorov–Sinai or topological)
$\tau(t)$	Information-theoretic time: $-\log \mu(C(a_{0:t-1}))$
T_m	Synchronization time
$\mathrm{Col}_u(m)$	Collision probability
$N(m,n)$	Capacity: max. simultaneously addressable events
$\prec_{\mathrm{prot}}$	Protocol causal precedence
$D_N^*(\alpha; x_0)$	Star discrepancy
$\kappa(A)$	Convergence coefficient: $A/\log \lambda_A$

Conventions.

Logarithms are natural (ln) unless otherwise stated. Entropy is measured in nats. The indicator function $\mathbf{1}_S$ takes value 1 on S and 0 elsewhere. For a probability measure ν and a measurable set A, we write $\nu(A)$ for the measure of A.

Appendix B Worked example: the Fibonacci chain

We instantiate the full framework on the simplest nontrivial case: the one-dimensional Fibonacci quasicrystal, obtained from a rank-2 CPS with golden-ratio slope.

B.1 CPS data

- Physical space $E = \mathbb{R}^1$, internal space $H = \mathbb{R}^1$.
- Lattice $\Gamma = \{(a + b\varphi,\ a + b\varphi') : a, b \in \mathbb{Z}\} \subset \mathbb{R}^2$, where $\varphi' = (1 - \sqrt{5})/2 = -1/\varphi$.
- Projections: $\pi_E(a + b\varphi, a + b\varphi') = a + b\varphi$, $\pi_I(a + b\varphi, a + b\varphi') = a + b\varphi'$.
- Window: $W = [-1/(2\varphi),\ 1/(2\varphi)]$, so $|W| = 1/\varphi$.

The model set $\Lambda(W) = \{a + b\varphi : (a, b) \in \mathbb{Z}^2,\ |a + b\varphi'| \leq 1/(2\varphi)\}$ is the classical Fibonacci point set. Since $\partial W = \{-1/(2\varphi), 1/(2\varphi)\}$ has measure zero under Haar measure on $\mathbb{R}^1$, Proposition 2.1 guarantees unique ergodicity of the associated hull dynamical system.

B.2 Readout and Sturmian coding

The internal-space dynamics reduces to a circle rotation $\theta \mapsto \theta + \varphi \pmod 1$ on $\mathbb{T}^1$. The readout map $R : \mathbb{T}^1 \to \{0, 1\}$ is defined by the Sturmian partition:

$$R(\theta) = \lfloor (t + 1)\varphi \rfloor - \lfloor t\varphi \rfloor,$$

where t indexes the scan step. The resulting symbolic sequence is the Fibonacci word $\omega_\varphi = (a_t)_{t \geq 0}$, an element of the golden-mean shift $X_\varphi \subset \{0, 1\}^{\mathbb{N}}$ (no two consecutive 1's).

Scan error.

For the box window W of length $L = 1/\varphi$, the Tanaka–Stokes identity (Theorem 3.3) gives

$$\varepsilon_m = \varepsilon_0 - \mathbb{E}[L_m^{1/2}],$$

where $L_m(\theta) = \sum_{t=0}^{m-1} \mathbf{1}_{\partial P}(\theta + t\varphi)$ counts boundary visits. Since ∂P consists of two points, the boundary cylinder dimension is $d_\partial = 0$ and Theorem 3.4 yields exponential convergence:

$$\varepsilon_m \leq K \cdot \varphi^{-m}.$$

This is the fastest possible exponential rate for a 1D CPS with algebraic slope (cf. Proposition 3.6).

B.3 Stabilization

Language size.

The golden-mean language has $|X_m| = F_{m+2}$ words. For $m = 5$: $|X_5| = F_7 = 13$ stabilized types out of $2^5 = 32$ raw words.

Fold map.

The Zeckendorf normalization $\mathrm{Fold}_5 : \{0,1\}^5 \to X_5$ acts by the rewrite rule $11 \mapsto 100$ applied left-to-right until no adjacent 1's remain. For example:

$$\mathrm{Fold}_5(11010) = \mathrm{Fold}_5(\underbrace{11}010) = 10110 \to 10\underbrace{11}0 \to 10100 \ \in X_5.$$

The confluence property (Theorem 4.2(i)) guarantees the result is independent of the order in which overlapping 11-patterns are resolved.

Fiber sizes.

The fiber $\mathcal{F}_5(x) = \mathrm{Fold}_5^{-1}(x)$ has size $d_5(x) = |\mathcal{F}_5(x)|$ depending on x. The total satisfies $\sum_{x \in X_5} d_5(x) = 2^5 = 32$, distributed over $|X_5| = 13$ types. The degeneracy histogram is a reproducible invariant (Proposition 7.1).

Gauge anomaly.

For $m = 5$, the worst-case gauge anomaly $G_5(\omega^*) = 5$ is achieved by the periodic sequence $\omega^* = (11111\cdots)$ restricted to length 5 (Theorem 4.3). The expected gauge anomaly density satisfies $\mathbb{E}[G_m]/m \to 4/9$ (Theorem 4.4).

B.4 Information-theoretic time

Under the Parry measure μ_φ on the golden-mean shift, the entropy rate is $h = \log \varphi \approx 0.4812$ nats. For a typical scan of length t, the information-theoretic time is

$$\tau(t) = -\log \mu_\varphi(C(a_{0:t-1})) \approx t \log \varphi,$$

with deviations controlled by the SMB theorem (Theorem 5.1).

The KL-ledger identity (Theorem 5.2) decomposes as:

$$H(\mu_m) = H(\pi_m) + \mathbb{E}_{\pi_m}[\log d_m] - D_{\mathrm{KL}}(\mu_m \| \mu_m^e).$$

For the Parry measure, $\mu = \mu^e$ (the Parry measure is already the maximum-entropy measure on X_φ), so $D_{\mathrm{KL}} = 0$ and the identity reduces to $H(\mu_m) = H(\pi_m) + \mathbb{E}[\log d_m]$.

B.5 Capacity and numerical verification

The double-budget capacity law (Theorem 6.1) for N simultaneously addressed events gives:

$$\log N \lesssim \min\{\varepsilon_m \cdot n,\ \tfrac{1}{2} h_2(u) \cdot m\},$$

where $\varepsilon_m \sim \varphi^{-m}$ (scan error) and $h_2(u)$ is the Rényi entropy of the address distribution.

TV certificate.

For the Fibonacci chain with slope $\alpha = \varphi - 1 = [0; 1, 1, 1, \dots]$, the TV certificate (Theorem 7.2) gives

$$D_{\mathrm{TV}}(\hat{\pi}_{m,N}, \pi_m^{\mathrm{fold}}) \leq (m+1)\frac{\log N + C}{N},$$

with $\kappa(1) = 1/\log\varphi \approx 2.078$, the unique minimum among all metallic means (Corollary 7.3).

Numerical protocol.

All quantities in this appendix are computed by `python3 scripts/run_all.py`, which outputs:

(i) diffraction spectrum (Bragg peaks on $\mathbb{Z}[\varphi]$);
(ii) scan error decay curve ε_m vs. m;
(iii) degeneracy histograms for $m = 3, 5, 8, 13$;
(iv) gauge anomaly distribution and density convergence to $4/9$;
(v) entropy-rate estimate $\hat{h}_N$ and convergence to $\log\varphi$;
(vi) TV distance $D_{\mathrm{TV}}(\hat{\pi}_{m,N}, \pi_m^{\mathrm{fold}})$ vs. N with certificate bound overlay.

All outputs are archived in `artifacts/` with parameter snapshots and random seeds.

References

[1] Meyer, Y.: Algebraic Numbers and Harmonic Analysis. North-Holland, Amsterdam (1972)

[2] Moody, R.V.: Model sets: a survey. In: Axel, F., Dénoyer, F., Gazeau, J.-P. (eds.) From Quasicrystals to More Complex Systems, pp. 145–166. Springer, Berlin (2000)

[3] Moody, R.V.: Meyer sets and their duals. In: Moody, R.V. (ed.) The Mathematics of Long-Range Aperiodic Order, pp. 403–441. Kluwer, Dordrecht (1997)

[4] Baake, M., Lenz, D.: Dynamical systems on translation bounded measures: pure point dynamical and diffraction spectra. Ergodic Theory Dynam. Systems **24**(6), 1867–1893 (2004)

[5] Baake, M., Grimm, U.: Aperiodic Order, Vol. 1: A Mathematical Invitation. Cambridge University Press, Cambridge (2013)

[6] Hof, A.: On diffraction by aperiodic structures. Comm. Math. Phys. **169**(1), 25–43 (1995)

[7] Richard, C., Strungaru, N.: Pure point diffraction and Poisson summation. Ann. Henri Poincaré **18**(12), 3903–3931 (2017)

[8] Schlottmann, M.: Generalized model sets and dynamical systems. In: Baake, M., Moody, R.V. (eds.) Directions in Mathematical Quasicrystals, pp. 143–159. AMS, Providence (2000)

[9] Lind, D., Marcus, B.: An Introduction to Symbolic Dynamics and Coding. Cambridge University Press, Cambridge (1995)

[10] Lothaire, M.: Algebraic Combinatorics on Words. Cambridge University Press, Cambridge (2002)

[11] Berthé, V., Yassawi, R.: Multidimensional continued fractions, substitutions, and Rauzy fractals. In: Berthoud, F. et al. (eds.) Substitution and Tiling Dynamics, Lecture Notes in Mathematics, vol. 2273, pp. 1–64. Springer, Cham (2020)

[12] Zeckendorf, E.: Représentation des nombres naturels par une somme de nombres de Fibonacci ou de nombres de Lucas. Bull. Soc. Roy. Sci. Liège **41**, 179–182 (1972)

[13] Khinchin, A.Ya.: Continued Fractions. University of Chicago Press, Chicago (1964)

[14] Cusick, T.W., Flahive, M.E.: The Markoff and Lagrange Spectra. AMS, Providence (1989)

[15] Cassels, J.W.S.: An Introduction to Diophantine Approximation. Cambridge University Press, Cambridge (1957)

[16] Kuipers, L., Niederreiter, H.: Uniform Distribution of Sequences. Wiley, New York (1974)

[17] Walters, P.: An Introduction to Ergodic Theory. Springer, New York (1982)

[18] Cover, T.M., Thomas, J.A.: Elements of Information Theory, 2nd edn. Wiley, Hoboken (2006)

[19] Breiman, L.: The individual ergodic theorem of information theory. Ann. Math. Statist. **28**(3), 809–811 (1957)

[20] Parry, W.: Intrinsic Markov chains. Trans. Amer. Math. Soc. **112**, 55–66 (1964)

[21] Connes, A., Rovelli, C.: Von Neumann algebra automorphisms and time-thermodynamics relation in generally covariant quantum theories. Classical Quantum Gravity **11**(12), 2899–2917 (1994)

[22] Rovelli, C.: Zakopane lectures on loop gravity. In: Proceedings of the 3rd Quantum Geometry and Quantum Gravity School. PoS(QGQGS 2011)003 (2011)

[23] Page, D.N., Wootters, W.K.: Evolution without evolution: dynamics described by stationary observables. Phys. Rev. D **27**(12), 2885–2892 (1983)

[24] Bombelli, L., Lee, J., Meyer, D., Sorkin, R.D.: Space-time as a causal set. Phys. Rev. Lett. **59**(5), 521–524 (1987)

[25] Sorkin, R.D.: Causal sets: discrete gravity. In: Gomberoff, A., Marolf, D. (eds.) Lectures on Quantum Gravity, pp. 305–327. Springer, Boston (2005)

[26] Henson, J.: The causal set approach to quantum gravity. In: Oriti, D. (ed.) Approaches to Quantum Gravity, pp. 393–413. Cambridge University Press, Cambridge (2009)

[27] Ashtekar, A., Lewandowski, J.: Background independent quantum gravity: a status report. Classical Quantum Gravity **21**(15), R53–R152 (2004)

[28] Thiemann, T.: Modern Canonical Quantum General Relativity. Cambridge University Press, Cambridge (2007)

[29] Pearl, J.: Causality: Models, Reasoning, and Inference, 2nd edn. Cambridge University Press, Cambridge (2009)

[30] Spirtes, P., Glymour, C., Scheines, R.: Causation, Prediction, and Search, 2nd edn. MIT Press, Cambridge (2000)

[31] Peters, J., Janzing, D., Schölkopf, B.: Elements of Causal Inference. MIT Press, Cambridge (2017)

[32] Rovelli, C.: Partial observables. Phys. Rev. D **65**(12), 124013 (2002)

[33] Dittrich, B.: Partial and complete observables for Hamiltonian constrained systems. Gen. Relativity Gravitation **39**(11), 1891–1927 (2007)

[34] Takesaki, M.: Theory of Operator Algebras II. Springer, Berlin (2003)

[35] Krieger, W.: On the subsystems of topological Markov chains. Ergodic Theory Dynam. Systems **2**(2), 195–202 (1982)

[36] Falconer, K.: Fractal Geometry: Mathematical Foundations and Applications, 2nd edn. Wiley, Chichester (2003)

[37] Lapidus, M.L., van Frankenhuijsen, M.: Fractal Geometry, Complex Dimensions and Zeta Functions. Springer, New York (2006)

[38] Regge, T.: General relativity without coordinates. Nuovo Cimento **19**(3), 558–571 (1961)

[39] Ambjørn, J., Görlich, A., Jurkiewicz, J., Loll, R.: Nonperturbative quantum gravity. Phys. Rep. **519**(4–5), 127–210 (2012)

[40] Loll, R.: Quantum gravity from causal dynamical triangulations: a review. Classical Quantum Gravity **37**(1), 013002 (2020)

[41] Connes, A.: Noncommutative Geometry. Academic Press, San Diego (1994)

[42] Kellendonk, J., Putnam, I.F.: The Ruelle–Sullivan map for actions of $\mathbb{R}^n$. Math. Ann. **334**(3), 693–711 (2006)

[43] Sadun, L.: Topology of Tiling Spaces. AMS, Providence (2008)

[44] Forrest, A., Hunton, J., Kellendonk, J.: Topological Invariants for Projection Method Patterns. Mem. Amer. Math. Soc. **159**(758) (2002)

[45] Senechal, M.: Quasicrystals and Geometry. Cambridge University Press, Cambridge (1995)

[46] de Bruijn, N.G.: Algebraic theory of Penrose's non-periodic tilings of the plane. Indag. Math. **43**(1), 39–66 (1981)

[47] Shechtman, D., Blech, I., Gratias, D., Cahn, J.W.: Metallic phase with long-range orientational order and no translational symmetry. Phys. Rev. Lett. **53**(20), 1951–1953 (1984)

[48] Levine, D., Steinhardt, P.J.: Quasicrystals: a new class of ordered structures. Phys. Rev. Lett. **53**(26), 2477–2480 (1984)

[49] Thuswaldner, J.M.: S-adic sequences: a bridge between dynamics, arithmetic, and geometry. In: Berthé, V., Rigo, M. (eds.) Sequences, Groups, and Number Theory, pp. 97–144. Birkhäuser, Cham (2018)

[50] Elser, V.: The diffraction pattern of projected structures. Acta Crystallogr. Sect. A **42**(1), 36–43 (1986)